



Approximation Algorithms for Big Data

Lecture 10: Low Rank Approximation

Spring 2026

Instructor: Tal Wagner

Administrative Announcements

- University-wide teaching survey is live:
- https://my.tau.ac.il/TAU_Student/TeachingSurvey
- Your feedback is highly valuable

הדקנט לחדשנות
בהוראה ובלמידה
אוניברסיטת תל אביב



סקר שביעות רצון מהוראה
סמסטר ב', תשפ"ז

יפתח מחר, 23/6 בשעה 8:00 למענה

ספרו לנו, איך עבר הסמסטר?



מקנה
נקודות
בבידינג

הקליקו לסקר

כניסה למידע האישי > מענה על הסקר

הסקר יימשך שלושה וחצי שבועות
ויהיה פתוח למענה עד ל-18/7/26

Administrative Announcements

- **Next week** (July 9): Guest lecture by **Dori Eldar** from **Mobileye**
- **Topic:** probabilistic framework for aligning multiple vehicle drives over a local map
- Details published on email and class forum on Moodle



The poster is for a "SPECIAL GUEST LECTURE" titled "AAA: Alignment, Assignment & Aggregation" with the subtitle "A Principled Approach to Multi-Drive Map Alignment". The speaker is Dori Eldar, Principal Engineer at Mobileye. The event is scheduled for July 9, 2026, at 15:00 in the Goldschlager Multidisciplinary Engineering Building, 3rd Floor, Room 315. The poster features the Mobileye logo and the Tel Aviv University logo at the top. The background is a blue gradient with a faint map-like pattern.

mobileye™

אוניברסיטת תל אביב
TEL AVIV UNIVERSITY

SPECIAL GUEST LECTURE

AAA: Alignment, Assignment
& Aggregation

A Principled Approach to Multi-Drive Map Alignment

 Dori Eldar
Principal Engineer, Mobileye

July 9, 2026 | 15:00
Goldschlager Multidisciplinary Engineering Building
3rd Floor, Room 315

Administrative Announcements

- Submission reminders (submit on Moodle):
 - Project proposals: by **July 11**
 - HW3: By **July 16**
 - Class engagement reporting: by **July 17**

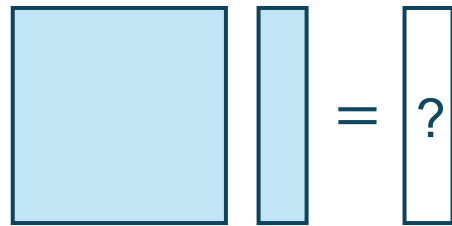
Plan

- Recap last time
- Today's topic: Approximate low-rank approximation via sketching
- Brief aside: low rank adaption (LoRA) in LLMs

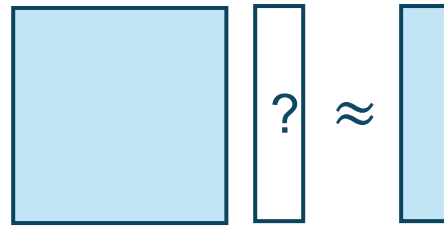
Recap

Computational Linear Algebra

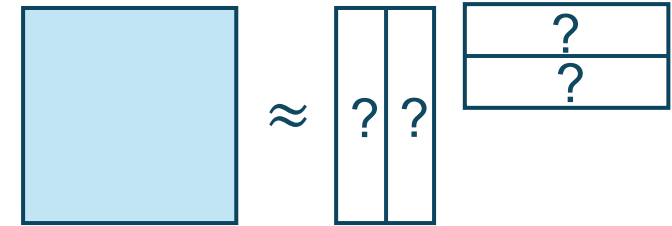
- Extremely useful and prevalent
- Extremely computationally taxing
- We will develop efficient randomized approximation algorithms in the spirit of class
- Why would the techniques we've seen so far be related?



Matrix-vector /
matrix-matrix
multiplication



Linear regression



Low-rank
approximation

AMM: Upshot

- We proved:
 - S with iid entries from $\frac{1}{\sqrt{m}} N(0,1)$ with $m = 1/\varepsilon^2 \delta$ satisfied 2-JLMP:

$$\sqrt{\mathbb{E}[(\|Sx\|_2^2 - 1)^2]} \leq \varepsilon\sqrt{\delta}$$

- 2-JLMP implies AMM:

$$\Pr[\|(SA)^T(SB) - A^T B\|_F \leq \varepsilon\|A\|_F\|B\|_F] \geq 1 - \delta$$

- Therefore: S with iid entries from $\frac{1}{\sqrt{m}} N(0,1)$ with $m = 1/\varepsilon^2 \delta$ satisfies AMM
- Remark: Can get AMM with only $m = \log\left(\frac{1}{\delta}\right)/\varepsilon^2$
 - Can be proved via higher JL moments

ℓ_2 -Regression

- Definition: in the ℓ_2 -regression problem,
 - Input: $A \in \mathbb{R}^{n \times d}$ and $b \in \mathbb{R}^d$
 - Output: $x^* = \min_{x \in \mathbb{R}^d} \|Ax - b\|_2$ (i.e., solve the linear equation as best as possible)
 - Exact solution? $x^* = (A^T A)^+ A^T b$
 - Time of exact solution?

- Approximate ℓ_2 -regression : Given ε, δ , find x such that

$$\Pr[\|Ax - b\|_2 \leq (1 + \varepsilon) \min_{x \in \mathbb{R}^d} \|Ax - b\|_2] \geq 1 - \delta$$

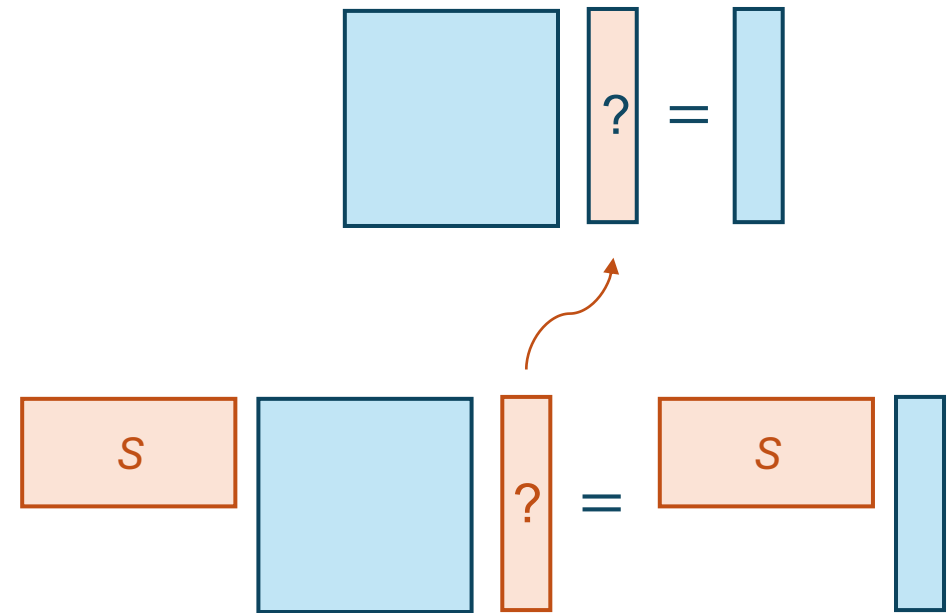
- Ideas how to achieve this?

ℓ_2 -Regression via Sketch-and-Solve

- How can we solve ℓ_2 -regression with an OSE?

- Sketch-and-solve:

- Sample S
- Solve $\tilde{x} = \min_{x \in \mathbb{R}^d} \|SAx - Sb\|_2$ exactly
- Return $\tilde{x}$



Oblivious Subspace Embedding (OSE)

- **Definition:** a distribution over random matrices $S \in \mathbb{R}^{m \times n}$ is an (ε, δ, d) -oblivious subspace embedding (OSE) if for $\mathbb{R}^n$ if for every given subspace $\mathcal{V} \subset \mathbb{R}^n$ of dimension d , we have:

$$\Pr[\forall x \in \mathcal{V}, \quad \|Sx\|_2^2 = (1 \pm \varepsilon)\|x\|_2^2] \geq 1 - \delta$$

- *"JL for an entire subspace"*
- **Equivalent definition:** S is an OSE if for every given matrix $A \in \mathbb{R}^{n \times d}$ we have:

$$\Pr[\forall x \in \mathbb{R}^n, \quad \|SAx\|_2^2 = (1 \pm \varepsilon)\|Ax\|_2^2] \geq 1 - \delta$$

- Why are these the same?

OSE $\Rightarrow$ Regression

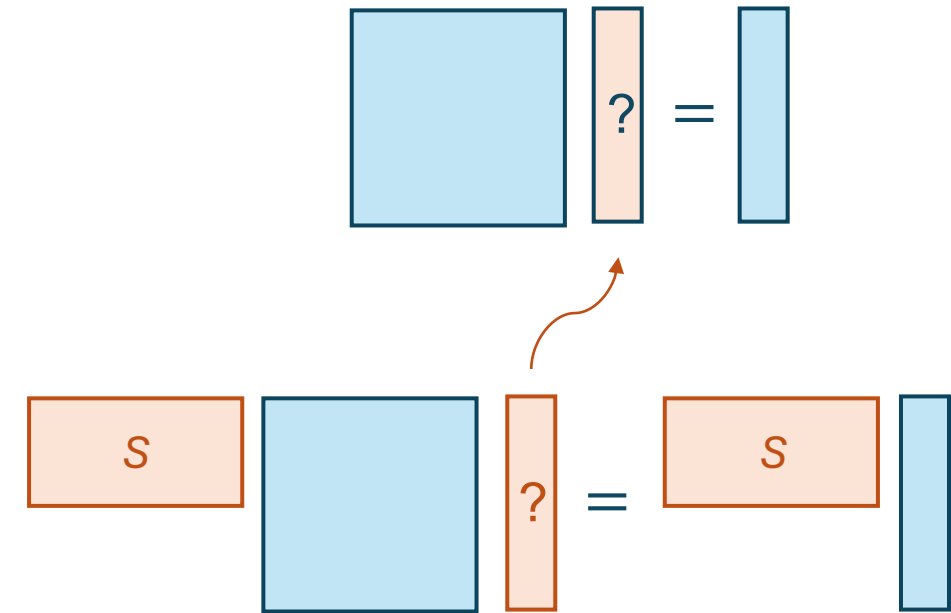
- How can we solve ℓ_2 -regression with an OSE?

- Sketch-and-solve:

- Sample S
- Solve $\tilde{x} = \min_{x \in \mathbb{R}^d} \|SAx - Sb\|_2$ exactly
- Return $\tilde{x}$
- **Observe:** if S is an OSE then

$$\|A\tilde{x} - b\|_2 \leq (1 + O(\varepsilon)) \min_{x \in \mathbb{R}^d} \|Ax - b\|_2 \quad (\text{why?})$$

- **Hence:** Our next goal is to build OSEs (ideas?)



JL => IPJL

- **Definition (“usual” JL):** a distribution over random matrices $S \in \mathbb{R}^{m \times n}$ satisfies the (ε, δ, N) -JL property if for every given set of N vectors $y_1, \dots, y_N \in \mathbb{R}^n$ we have:

$$\Pr \left[\forall i, j: \|Sy_i - Sy_j\|_2 = (1 \pm \varepsilon) \|y_i - y_j\|_2 \right] \geq 1 - \delta$$

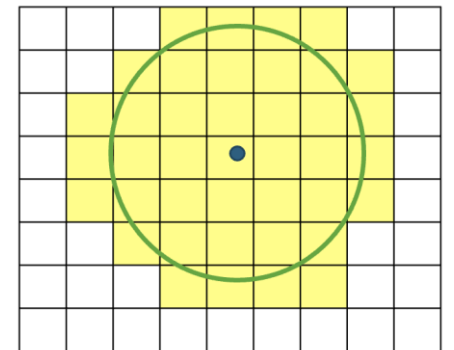
- **Definition:** $S \in \mathbb{R}^{m \times n}$ satisfies the (ε, δ, N) -inner product JL property (IPJL) if for every given set of N vectors $x_1, \dots, x_N \in \mathbb{R}^n$ we have:

$$\Pr \left[\forall i, j: |x_i^T S^T S x_j - x_i^T x_j| < \varepsilon \|x_i\|_2 \|x_j\|_2 \right] \geq 1 - \delta$$

- **Claim:** $(\varepsilon, \delta, N^2)$ -JL implies $(O(\varepsilon), \delta, N)$ -IPJL
 - **Proof?** The N^2 vectors are every x_i and every $x_i + x_j$ (why?)
- **Corollary:** $S \in \mathbb{R}^{m \times n}$ with iid Gaussians and $m = O\left(\frac{\log(N/\delta)}{\varepsilon^2}\right)$ satisfies (ε, δ, N) -IPJL

IPJL $\Rightarrow$ OSE

- Let S be a random matrix satisfying (ε, δ, N) -IPJL
- Let $\mathcal{V} \subset \mathbb{R}^n$ be a subspace of dimension d
- We want to show: w.p. $1 - \delta$, $\forall x \in \mathcal{V}$, $\|Sx\|_2^2 = (1 \pm \varepsilon)\|x\|_2^2$
- It suffices to show this for all unit vectors $x \in \mathcal{V}$ (why?)
- **Definition:** an η -net for the unit sphere in $\mathcal{V}$ is a subset $Z \subset \mathcal{V}$ such that for every unit vector $x \in \mathcal{V}$, there is $z \in Z$ such that $\|x - z\|_2 \leq \eta$.
- **Lemma:** The unit sphere in $\mathcal{V}$ has an η -net of size $|Z| \leq O\left(\frac{1}{\eta}\right)^d$.
 - **Proof?** We've seen this!
 - We will take $\eta = 1/2$, so $|Z| = O(1)^d$



OSE => Sketch-and-Solve Regression

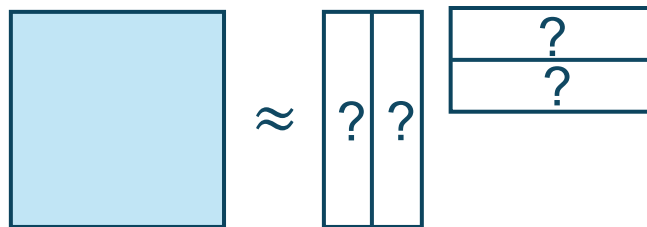
- **Corollary:**

- Let $A \in \mathbb{R}^{n \times d}$, $b \in \mathbb{R}^d$ be a regression problem.
- Let $m = \tilde{O}(d/\varepsilon^2)$.
- Let $S \in \mathbb{R}^{m \times n}$ be a random matrix with iid Gaussian entries $\frac{1}{\sqrt{m}} N(0,1)$.
- Then, the sketch-and-solve solution $\tilde{x} = \min_{x \in \mathbb{R}^d} \|SAx - Sb\|_2$ satisfies the approximate regression guarantee:

$$\|A\tilde{x} - b\|_2 \leq (1 + O(\varepsilon)) \min_{x \in \mathbb{R}^d} \|Ax - b\|_2$$

- What is the running time?
- See a big problem here?
- See a way around it?

Approximate Low Rank Approximation



Matrix Low Rank Approximation (LRA) (1/2)

- **Definition** – matrix low-rank approximation:

- Input: matrix $A \in \mathbb{R}^{n \times d}$ of arbitrary rank, and target rank $k \ll \min\{n, d\}$
- Goal: Find a matrix $A' \in \mathbb{R}^{n \times d}$ of rank at most k closest to A in some metric

$$A' = \operatorname{argmin}_{\operatorname{rank}(A') \leq k} \|A - A'\|$$

- **What metric?**

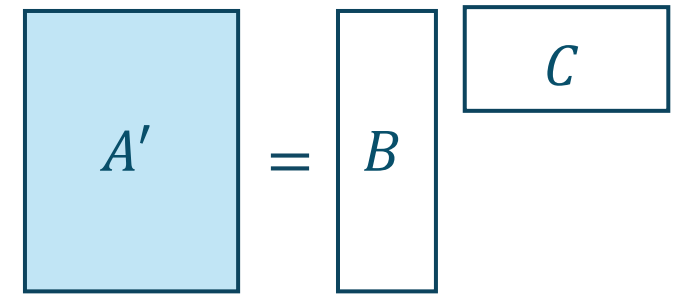
- Common choices: Frobenius norm $\|A - A'\|_F$, spectral norm $\|A - A'\|_2$
- We will focus on Frobenius norm

Matrix Low Rank Approximation (LRA) (2/2)

- Every $A' \in \mathbb{R}^{n \times d}$ of rank k can be written as $A' = BC$ where $B \in \mathbb{R}^{n \times k}$ and $C \in \mathbb{R}^{k \times d}$
 - How?

- Therefore, low-rank approximation is equivalent to:

$$B, C = \operatorname{argmin}_{B \in \mathbb{R}^{n \times k}, C \in \mathbb{R}^{k \times d}} \|A - BC\|$$


$$A' = B C$$

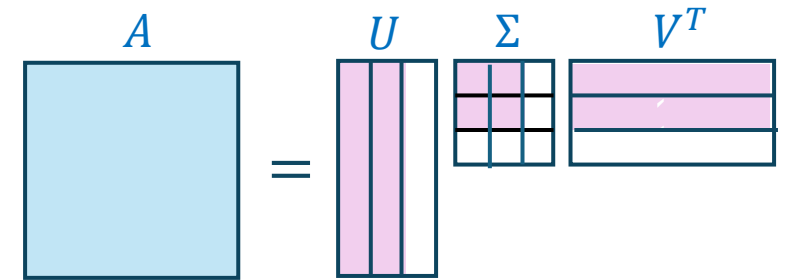
- Why LRA?
 - How much space it takes to store A vs. B, C ?
 - Given $x \in \mathbb{R}^d$, how much time it takes to compute Ax vs. BCx ?

Optimal LRA via SVD (Eckhart-Young-Mirsky Theorem)

- **Recall – SVD:** Every $A \in \mathbb{R}^{n \times d}$ of rank k can be written as $A = U\Sigma V^T$ such that:

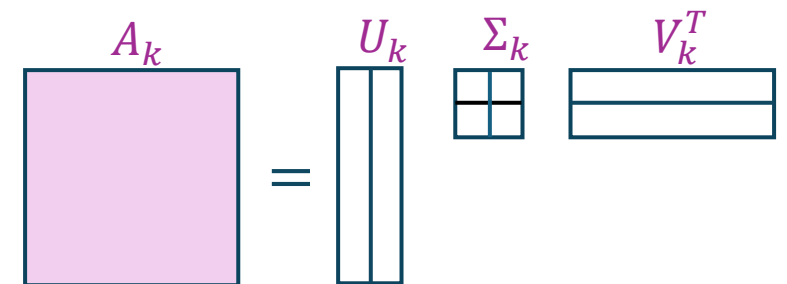
- $U \in \mathbb{R}^{n \times k}$ has orthonormal columns
- $V \in \mathbb{R}^{d \times k}$ has orthonormal columns
- $\Sigma \in \mathbb{R}^{k \times k}$ is diagonal with sorted entries :

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k > 0$$



- Equivalent writing: $A = \sum_{i=1}^r \sigma_i \cdot u_i v_i^T$
- Let $k < r$ be the target low rank
- Form A_k by taking the first r rows/cols of U, Σ, V

$$A_k = \sum_{i=1}^k \sigma_i \cdot u_i v_i^T \text{ (sum is only up to } k \text{)}$$



- **Theorem:** A_k is an optimal rank- r LRA for A in both Frobenius and spectral norm
 - Intuition?

Approximate LRA

- We just saw: optimal LRA can be computed via SVD
- Consequences:
 - Optimal LRA can be computed in polynomial time (what's the running time?)
 - Optimal rank- k LRA has Frobenius error $\|A - A_k\|_F$

- **Definition – randomized approximate LRA:**

- Input: matrix $A \in \mathbb{R}^{n \times d}$ of arbitrary rank, and target rank $k \ll \min\{n, d\}$
- Goal: Find $A' \in \mathbb{R}^{n \times d}$ that with probability $1 - \delta$ satisfies:

$$\|A - A'\|_F \leq (1 + \varepsilon)\|A - A_k\|_F$$

- **Our goal**: Approx LRA algorithm in time $O(nd \cdot (k/\varepsilon)^{O(1)})$ (linear in the size of A)

Approximate LRA via Sketching

- Let $A \in \mathbb{R}^{n \times d}$ and target rank k be the inputs to LRA
- Let $S \in \mathbb{R}^{m \times n}$ be a sketching matrix, where we will choose $k \ll m \ll n$
- What does “sketch and solve” mean in LRA? (Can we just return an LRA of SA ?)
- Let $A' \in \mathbb{R}^{n \times d}$ have rank k
- Then, the rows of A' lie in a subspace of $\mathbb{R}^n$ of dimension k (why?)
- **Exact LRA** can choose the best rows for A' from all of $\mathbb{R}^n$
- **Exact LRA** will choose the best rows from A' from the row-space of A (why?)
- **Approximate LRA** will choose the best rows from A' from the row-space of SA
 - Intuition: $\dim(SA) = m$
 - Since $m \gg k$, we have enough richness from which to choose rows for A'
 - Since $m \ll n$, choosing the best rows for A' could be efficient

Approximate LRA via Sketching

- **Algorithm “plan”:**

- Sample a sketching matrix $S \in \mathbb{R}^{m \times n}$
- Compute SA
- Compute and return the best rank- k LRA of A within the row-space of SA :

$$A' = \operatorname{argmin}_{A': \text{rows}(A') \subset \text{span}(\text{rows}(SA))} \|A - A'\|_F$$

- **Claim:** Every $A' \in \mathbb{R}^{n \times d}$ of rank k in the row-space of SA is equal to $A' = BSA$ where $B \in \mathbb{R}^{n \times m}$ is some matrix of rank k
- **Hence:** we are looking for the best $B \in \mathbb{R}^{n \times m}$ of rank k :

$$A' = BSA = \operatorname{argmin}_{B \in \mathbb{R}^{n \times m}, \text{rank}(B)=k} \|A - BSA\|_F$$

Approximate LRA via Sketching

- **Algorithm “plan”:**

- Sample a sketching matrix $S \in \mathbb{R}^{m \times n}$
- Compute SA
- Compute and return the best rank- k LRA of A within the row-space of SA :

$$A' = \operatorname{argmin}_{A': \text{rows}(A') \subset \text{span}(\text{rows}(SA))} \|A - A'\|_F$$

- **Claim:** Every $A' \in \mathbb{R}^{n \times d}$ of rank k in the row-space of SA is equal to $A' = BSA$ where $B \in \mathbb{R}^{n \times m}$ is some matrix of rank k
- **Hence:** we are looking for the best $B \in \mathbb{R}^{n \times m}$ of rank k :

$$A' = BSA = \operatorname{argmin}_{B \in \mathbb{R}^{n \times m}, \text{rank}(B)=k} \|A - BSA\|_F$$

- **What are the two big gaps we need to fill?**

- **Prove efficiency:** Given SA , how can we find the best B efficiently?
- **Prove accuracy:** How do we show the best B satisfies approximate LRA:

$$\Pr[\|A - A'\|_F \leq (1 + \varepsilon)\|A - A_k\|_F] \geq 1 - \delta ?$$

Best LRA in Subspace

- Let $SA = U_S \Sigma_S V_S^T$ be the SVD of SA
 - **Note:** This is the SVD of SA , not S
- Observe: the row-space of SA is equal to the row-space of V_S^T (why?)
- Therefore: We are looking for C that minimizes $\|A - CV_S^T\|_F$
- **Theorem**: The best C is $C = [AV_S]_k$.
- Hence: The best rank- k LRA of A within the row-space of SA is $A' = [AV_S]_k V_S^T$.

- **This gives us our concrete LRA-via-sketching algorithm:**

1. Sample sketching matrix $S \in \mathbb{R}^{m \times n}$
2. Compute the SVD $SA = U_S \Sigma_S V_S^T$ of SA
3. Compute $[AV_S]_k$
4. Return $A' = [AV_S]_k V_S^T$

Running
time?

Best LRA in Subspace: Proof

- **Theorem:** The best B is $C = [AV_S]_k$.
- We need to show: For all C of rank k , $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$.

Best LRA in Subspace: Proof

- **Theorem:** The best B is $C = [AV_S]_k$.
- We need to show: For all C of rank k , $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$.
- **Proof:**

Best LRA in Subspace: Proof

- **Theorem:** The best B is $C = [AV_S]_k$.
- We need to show: For all C of rank k , $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$.
- **Proof:**
 - **Claim:** For every matrix M , we have $\|M\|_F = \|MV_S^T\|_F$. (Proof?)

Best LRA in Subspace: Proof

- **Theorem:** The best B is $C = [AV_S]_k$.
- We need to show: For all C of rank k , $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$.
- **Proof:**
 - **Claim:** For every matrix M , we have $\|M\|_F = \|MV_S^T\|_F$. (Proof?)
 - Let $C \in \mathbb{R}^{n \times m}$ have rank k .

Best LRA in Subspace: Proof

- **Theorem:** The best B is $C = [AV_S]_k$.
- We need to show: For all C of rank k , $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$.
- **Proof:**
 - **Claim:** For every matrix M , we have $\|M\|_F = \|MV_S^T\|_F$. (Proof?)
 - Let $C \in \mathbb{R}^{n \times m}$ have rank k .
 - We have:
$$\underbrace{\|AV_S V_S^T - [AV_S]_k V_S^T\|_F}_{\text{Claim}} = \underbrace{\|AV_S - [AV_S]_k\|_F}_{\text{Why?}} \leq \underbrace{\|AV_S - C\|_F}_{\text{Claim}} = \|AV_S V_S^T - CV_S^T\|_F$$

Best LRA in Subspace: Proof

- **Theorem:** The best B is $C = [AV_S]_k$.

- We need to show: For all C of rank k , $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$.

- **Proof:**

- **Claim:** For every matrix M , we have $\|M\|_F = \|MV_S^T\|_F$. (Proof?)

- Let $C \in \mathbb{R}^{n \times m}$ have rank k .

- We have:

$$\underbrace{\|AV_S V_S^T - [AV_S]_k V_S^T\|_F}_{\text{Claim}} = \underbrace{\|AV_S - [AV_S]_k\|_F}_{\text{Why?}} \leq \underbrace{\|AV_S - C\|_F}_{\text{Claim}} = \|AV_S V_S^T - CV_S^T\|_F$$

- Hence:

- $\|A - [AV_S]_k V_S^T\|_F = \|(A - [AV_S]_k V_S^T)V_S V_S^T\|_F + \|(A - [AV_S]_k V_S^T)(I - V_S V_S^T)\|_F$
- $= \|AV_S V_S^T - [AV_S]_k V_S^T\|_F + \|A - AV_S V_S^T\|_F$
- $\leq \|AV_S V_S^T - CV_S^T\|_F + \|A - AV_S V_S^T\|_F$
- $= \|A - CV_S^T\|_F.$



Best LRA in Subspace: Proof

- **Theorem:** The best B is $C = [AV_S]_k$.
- We need to show: For all C of rank k , $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$.

- **Proof:**

- **Claim:** For every matrix M , we have $\|M\|_F = \|MV_S^T\|_F$. (Proof?)

- Let $C \in \mathbb{R}^{n \times m}$ have rank k .

- We have:

$$\|AV_S V_S^T - [AV_S]_k V_S^T\|_F = \|AV_S - [AV_S]_k\|_F \leq \|AV_S - C\|_F = \|AV_S V_S^T - CV_S^T\|_F$$

Claim
Why?
Claim

- Hence:

- $\|A - [AV_S]_k V_S^T\|_F = \|(A - [AV_S]_k V_S^T)V_S V_S^T\|_F + \|(A - [AV_S]_k V_S^T)(I - V_S V_S^T)\|_F$
- $= \|AV_S V_S^T - [AV_S]_k V_S^T\|_F + \|A - AV_S V_S^T\|_F$
- $\leq \|AV_S V_S^T - CV_S^T\|_F + \|A - AV_S V_S^T\|_F$
- $= \|A - CV_S^T\|_F$.

???



Best LRA in Subspace: Proof

- **Theorem:** The best B is $C = [AV_S]_k$.

- We need to show: For all C of rank k , $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$.

- **Proof:**

- **Claim:** For every matrix M , we have $\|M\|_F = \|MV_S^T\|_F$. (Proof?)

- Let $C \in \mathbb{R}^{n \times m}$ have rank k .

- We have:

$$\|AV_S V_S^T - [AV_S]_k V_S^T\|_F = \|AV_S - [AV_S]_k\|_F \leq \|AV_S - C\|_F = \|AV_S V_S^T - CV_S^T\|_F$$

Claim
Why?
Claim

- Hence:

$$\begin{aligned} \|A - [AV_S]_k V_S^T\|_F &= \|(A - [AV_S]_k V_S^T) V_S V_S^T\|_F + \|(A - [AV_S]_k V_S^T)(I - V_S V_S^T)\|_F \\ &= \|AV_S V_S^T - [AV_S]_k V_S^T\|_F + \|A - AV_S V_S^T\|_F \\ &\leq \|AV_S V_S^T - CV_S^T\|_F + \|A - AV_S V_S^T\|_F \\ &= \|A - CV_S^T\|_F. \end{aligned}$$

Pythagorean
identity



LRA via Sketching: Status So Far

LRA via Sketching: Status So Far

- We proved **efficiency**:
 - Our strategy: Find the best rank- k LRA for A in the row-space of SA

LRA via Sketching: Status So Far

- We proved **efficiency**:
 - Our strategy: Find the best rank- k LRA for A in the row-space of SA
 - Equivalently: Find $B \in \mathbb{R}^{n \times m}$ of rank k that minimizes $\|A - BSA\|_F$

LRA via Sketching: Status So Far

- We proved **efficiency**:
 - Our strategy: Find the best rank- k LRA for A in the row-space of SA
 - Equivalently: Find $B \in \mathbb{R}^{n \times m}$ of rank k that minimizes $\|A - BSA\|_F$
 - Equivalently: Finding the C of rank k that minimizes $\|A - CV_S^T\|_F$

LRA via Sketching: Status So Far

- We proved **efficiency**:
 - Our strategy: Find the best rank- k LRA for A in the row-space of SA
 - Equivalently: Find $B \in \mathbb{R}^{n \times m}$ of rank k that minimizes $\|A - BSA\|_F$
 - Equivalently: Finding the C of rank k that minimizes $\|A - CV_S^T\|_F$
 - This has a closed form solution: $C^* = [AV_S]_k$, where $SA = U_S \Sigma_S V_S^T$ is the SVD of SA .

LRA via Sketching: Status So Far

- We proved **efficiency**:
 - Our strategy: Find the best rank- k LRA for A in the row-space of SA
 - Equivalently: Find $B \in \mathbb{R}^{n \times m}$ of rank k that minimizes $\|A - BSA\|_F$
 - Equivalently: Finding the C of rank k that minimizes $\|A - CV_S^T\|_F$
 - This has a closed form solution: $C^* = [AV_S]_k$, where $SA = U_S \Sigma_S V_S^T$ is the SVD of SA .
 - Thus, we arrived at the **efficient approximate LRA algorithm**:
 - Compute the SVD $SA = U \Sigma V_S^T$
 - Return $A' = [AV_S]_k V_S^T$.

LRA via Sketching: Status So Far

- We proved **efficiency**:
 - Our strategy: Find the best rank- k LRA for A in the row-space of SA
 - Equivalently: Find $B \in \mathbb{R}^{n \times m}$ of rank k that minimizes $\|A - BSA\|_F$
 - Equivalently: Finding the C of rank k that minimizes $\|A - CV_S^T\|_F$
 - This has a closed form solution: $C^* = [AV_S]_k$, where $SA = U_S \Sigma_S V_S^T$ is the SVD of SA .
 - Thus, we arrived at the **efficient approximate LRA algorithm**:
 - Compute the SVD $SA = U \Sigma V_S^T$
 - Return $A' = [AV_S]_k V_S^T$.
- Next, we need to show **accuracy**: For a good choice of random sketching matrix S ,
$$\Pr[\|A - [AV]_k V^T\|_F \leq (1 + \varepsilon) \|A - A_k\|_F] \geq 1 - \delta.$$

LRA via Sketching: Accuracy

LRA via Sketching: Accuracy

- Recall AMM: $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM if for every pair of matrices $A, B \in \mathbb{R}^{n \times d}$,
$$\Pr[\|(SA)^T(SB) - A^T B\|_F \leq \varepsilon \|A\|_F \|B\|_F] \geq 1 - \delta$$

LRA via Sketching: Accuracy

- Recall AMM: $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM if for every pair of matrices $A, B \in \mathbb{R}^{n \times d}$,

$$\Pr[\|(SA)^T(SB) - A^T B\|_F \leq \varepsilon \|A\|_F \|B\|_F] \geq 1 - \delta$$

- Theorem:** Let $A \in \mathbb{R}^{n \times d}$ and target rank k be the input to LRA. If $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM, then $A' = [AV_S]_k V_S^T$ is an $(k^4 \varepsilon, \delta)$ -approximate LRA for A :

$$\Pr[\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2] > 1 - \delta$$

LRA via Sketching: Accuracy

- Recall AMM: $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM if for every pair of matrices $A, B \in \mathbb{R}^{n \times d}$,

$$\Pr[\|(SA)^T(SB) - A^T B\|_F \leq \varepsilon \|A\|_F \|B\|_F] \geq 1 - \delta$$

- Theorem:** Let $A \in \mathbb{R}^{n \times d}$ and target rank k be the input to LRA. If $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM, then $A' = [AV_S]_k V_S^T$ is an $(k^4 \varepsilon, \delta)$ -approximate LRA for A :

$$\Pr[\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2] > 1 - \delta$$

- And A' can be computed in time $O(ndm^2)$.

LRA via Sketching: Accuracy

- Recall AMM: $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM if for every pair of matrices $A, B \in \mathbb{R}^{n \times d}$,

$$\Pr[\|(SA)^T(SB) - A^T B\|_F \leq \varepsilon \|A\|_F \|B\|_F] \geq 1 - \delta$$

- Theorem:** Let $A \in \mathbb{R}^{n \times d}$ and target rank k be the input to LRA. If $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM, then $A' = [AV_S]_k V_S^T$ is an $(k^4 \varepsilon, \delta)$ -approximate LRA for A :

$$\Pr[\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2] > 1 - \delta$$

- And A' can be computed in time $O(ndm^2)$.
- Recall we saw (ε, δ) -AMM with $m \approx 1/\varepsilon^2$ rows (ignoring δ)

LRA via Sketching: Accuracy

- Recall AMM: $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM if for every pair of matrices $A, B \in \mathbb{R}^{n \times d}$,

$$\Pr[\|(SA)^T(SB) - A^T B\|_F \leq \varepsilon \|A\|_F \|B\|_F] \geq 1 - \delta$$

- Theorem:** Let $A \in \mathbb{R}^{n \times d}$ and target rank k be the input to LRA. If $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM, then $A' = [AV_S]_k V_S^T$ is an $(k^4 \varepsilon, \delta)$ -approximate LRA for A :

$$\Pr[\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2] > 1 - \delta$$

- And A' can be computed in time $O(ndm^2)$.
- Recall we saw (ε, δ) -AMM with $m \approx 1/\varepsilon^2$ rows (ignoring δ)
- Corollary:** (ε, δ) -approximate LRA in time $O(nd \cdot (k/\varepsilon)^{O(1)})$.
 - Linear in the size of A

LRA Accuracy: Proof

- **Theorem:** If $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM, then $A' = [AV_S]_k V_S^T$ satisfies
$$\Pr[\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2] > 1 - \delta$$

LRA Accuracy: Proof

- **Theorem:** If $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM, then $A' = [AV_S]_k V_S^T$ satisfies

$$\Pr[\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2] > 1 - \delta$$

- **Proof:**

- Let $A = U\Sigma V^T$ be the SVD of A

$$A = U \Sigma V^T$$

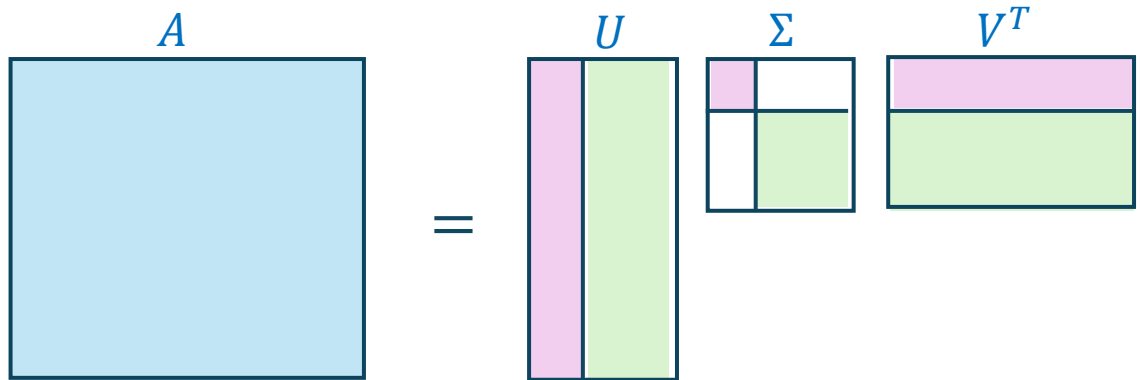
LRA Accuracy: Proof

- **Theorem:** If $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM, then $A' = [AV_S]_k V_S^T$ satisfies

$$\Pr[\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2] > 1 - \delta$$

- **Proof:**

- Let $A = U\Sigma V^T$ be the SVD of A
- We break it into the **top- k space** and the **rest**:
 - U_k = first k columns of U
 - U_{-k} = rest of columns of U
 - V_k^T = first k rows of V
 - V_{-k}^T = rest of rows of V
 - $\Sigma = \begin{bmatrix} \Sigma_k & 0 \\ 0 & \Sigma_{-k} \end{bmatrix}$



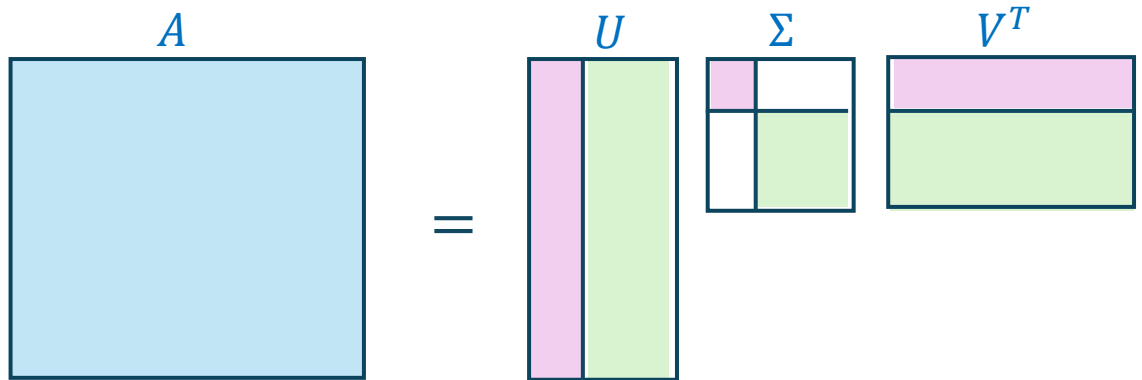
LRA Accuracy: Proof

- **Theorem:** If $S \in \mathbb{R}^{n \times m}$ satisfies (ε, δ) -AMM, then $A' = [AV_S]_k V_S^T$ satisfies

$$\Pr[\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2] > 1 - \delta$$

- **Proof:**

- Let $A = U\Sigma V^T$ be the SVD of A
- We break it into the **top- k space** and the **rest**:
 - U_k = first k columns of U
 - U_{-k} = rest of columns of U
 - V_k^T = first k rows of V
 - V_{-k}^T = rest of rows of V
 - $\Sigma = \begin{bmatrix} \Sigma_k & 0 \\ 0 & \Sigma_{-k} \end{bmatrix}$



Observe:

$$A_k = U_k \Sigma_k V_k^T$$

$$A - A_k = U_{-k} \Sigma_{-k} V_{-k}^T$$

LRA Accuracy: Proof

- Recall we proved: $A' = [AV_S]_k V_S^T$ is the best rank- k of A within the row-space of SA

LRA Accuracy: Proof

- Recall we proved: $A' = [AV_S]_k V_S^T$ is the best rank- k of A within the row-space of SA
- Meaning: $A' = B^* SA$ where $B^* = \operatorname{argmin}_{\operatorname{rank}(B)=k} \|A - BSA\|_F$

LRA Accuracy: Proof

- Recall we proved: $A' = [AV_S]_k V_S^T$ is the best rank- k of A within the row-space of SA
- Meaning: $A' = B^* SA$ where $B^* = \operatorname{argmin}_{\operatorname{rank}(B)=k} \|A - BSA\|_F$
- Thus: to prove $\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$, it suffices to find any B of rank k that satisfies

$$\|A - BSA\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$$

LRA Accuracy: Proof

- Recall we proved: $A' = [AV_S]_k V_S^T$ is the best rank- k of A within the row-space of SA
- Meaning: $A' = B^* SA$ where $B^* = \operatorname{argmin}_{\operatorname{rank}(B)=k} \|A - BSA\|_F$
- Thus: to prove $\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$, it suffices to find any B of rank k that satisfies

$$\|A - BSA\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$$

- We choose:

$$B := U_k (S U_k)^+ = U_k (U_k^T S^T S U_k)^{-1} U_k^T S^T$$

LRA Accuracy: Proof

- Recall we proved: $A' = [AV_S]_k V_S^T$ is the best rank- k of A within the row-space of SA
- Meaning: $A' = B^* SA$ where $B^* = \operatorname{argmin}_{\operatorname{rank}(B)=k} \|A - BSA\|_F$
- Thus: to prove $\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$, it suffices to find any B of rank k that satisfies

$$\|A - BSA\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$$

- We choose:

$$B := U_k (S U_k)^+ = U_k (U_k^T S^T S U_k)^{-1} U_k^T S^T$$

- For brevity we denote $Z = U_k^T S^T S U_k$, hence $B = U_k Z^{-1} U_k^T S^T$

LRA Accuracy: Proof

- Recall we proved: $A' = [AV_S]_k V_S^T$ is the best rank- k of A within the row-space of SA
- Meaning: $A' = B^* SA$ where $B^* = \operatorname{argmin}_{\operatorname{rank}(B)=k} \|A - BSA\|_F$
- Thus: to prove $\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$, it suffices to find any B of rank k that satisfies

$$\|A - BSA\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$$

- We choose:

$$B := U_k (S U_k)^+ = U_k (U_k^T S^T S U_k)^{-1} U_k^T S^T$$

- For brevity we denote $Z = U_k^T S^T S U_k$, hence $B = U_k Z^{-1} U_k^T S^T$
- Note: we have assumed Z is invertible; we will soon prove this is true

Interlude for Intuition: Why $B := U_k(SU_k)^+$?

Intuition for the choice of B in the proof (this is not part of the formal proof):

Interlude for Intuition: Why $B := U_k(SU_k)^+$?

Intuition for the choice of B in the proof (this is not part of the formal proof):

- Let a be a column of A
- Intuitively, we want the best approximation of a in the top- k column subspace of A

Interlude for Intuition: Why $B := U_k(SU_k)^+$?

Intuition for the choice of B in the proof (this is not part of the formal proof):

- Let a be a column of A
- Intuitively, we want the best approximation of a in the top- k column subspace of A

Interlude for Intuition: Why $B := U_k(SU_k)^+$?

Intuition for the choice of B in the proof (this is not part of the formal proof):

- Let a be a column of A
- Intuitively, we want the best approximation of a in the top- k column subspace of A
- That is, we want to solve **(1)**: $\min_x \|U_k x - a\|_2$

Interlude for Intuition: Why $B := U_k(SU_k)^+$?

Intuition for the choice of B in the proof (this is not part of the formal proof):

- Let a be a column of A
- Intuitively, we want the best approximation of a in the top- k column subspace of A
- That is, we want to solve **(1)**: $\min_x \|U_k x - a\|_2$
- The sketch-and-solve analog under S is **(2)**: $\min_x \|SU_k x - Sa\|_2$

Interlude for Intuition: Why $B := U_k(SU_k)^+$?

Intuition for the choice of B in the proof (this is not part of the formal proof):

- Let a be a column of A
- Intuitively, we want the best approximation of a in the top- k column subspace of A
- That is, we want to solve **(1)**: $\min_x \|U_k x - a\|_2$
- The sketch-and-solve analog under S is **(2)**: $\min_x \|SU_k x - Sa\|_2$
- The optimal min-length solution to **(2)** is: $x_S^* = (SU_k)^+ Sa$

Interlude for Intuition: Why $B := U_k(SU_k)^+$?

Intuition for the choice of B in the proof (this is not part of the formal proof):

- Let a be a column of A
- Intuitively, we want the best approximation of a in the top- k column subspace of A
- That is, we want to solve **(1)**: $\min_x \|U_k x - a\|_2$
- The sketch-and-solve analog under S is **(2)**: $\min_x \|SU_k x - Sa\|_2$
- The optimal min-length solution to **(2)** is: $x_S^* = (SU_k)^+ Sa$
- Plugging it as the approximation for a in **(1)**, we get: $U_k x_S^* = U_k (SU_k)^+ Sa = B Sa$

Interlude for Intuition: Why $B := U_k(SU_k)^+$?

Intuition for the choice of B in the proof (this is not part of the formal proof):

- Let a be a column of A
- Intuitively, we want the best approximation of a in the top- k column subspace of A
- That is, we want to solve **(1)**: $\min_x \|U_k x - a\|_2$
- The sketch-and-solve analog under S is **(2)**: $\min_x \|SU_k x - Sa\|_2$
- The optimal min-length solution to **(2)** is: $x_S^* = (SU_k)^+ Sa$
- Plugging it as the approximation for a in **(1)**, we get: $U_k x_S^* = U_k (SU_k)^+ Sa = B Sa$
- Applying this to each column of A :

With this B , BSA is the "best approx" of A in its top- k space under sketching with S

Back to LRA Accuracy Proof

- Recall we proved: $A' = [AV_S]_k V_S^T$ is the best rank- k of A within the row-space of SA
- Meaning: $A' = B^* SA$ where $B^* = \operatorname{argmin}_{\operatorname{rank}(B)=k} \|A - BSA\|_F$
- Thus: to prove $\|A - A'\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$, it suffices to find any B of rank k that satisfies

$$\|A - BSA\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$$

- We choose:

$$B := U_k (S U_k)^+ = U_k (U_k^T S^T S U_k)^{-1} U_k^T S^T$$

- For brevity we denote $Z = U_k^T S^T S U_k$, hence $B = U_k Z^{-1} U_k^T S^T$
- Note: we have assumed Z is invertible; we will soon prove this is true

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|BSA - A\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|BSA - A\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$
- We have:
 - $\|BSA - A\|_F^2 = \|U_k Z^{-1} U_k^T S^T S A - A\|_F^2$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|BSA - A\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$

- We have:

$$\begin{aligned} \|BSA - A\|_F^2 &= \|U_k Z^{-1} U_k^T S^T S A - A\|_F^2 \\ &= \|\underbrace{U_k U_k^T}_{???} (U_k Z^{-1} U_k^T S^T S A - A)\|_F^2 + \|(I - U_k U_k^T) (U_k Z^{-1} U_k^T S^T S A - A)\|_F^2 \end{aligned}$$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|BSA - A\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$

- We have:

$$\begin{aligned} \|BSA - A\|_F^2 &= \|U_k Z^{-1} U_k^T S^T S A - A\|_F^2 \\ &= \|U_k U_k^T (U_k Z^{-1} U_k^T S^T S A - A)\|_F^2 + \|(I - U_k U_k^T)(U_k Z^{-1} U_k^T S^T S A - A)\|_F^2 \end{aligned}$$

Pythagorean
identity with
 $U_k U_k^T$



LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|BSA - A\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$

- We have:

$$\begin{aligned} \bullet \quad \|BSA - A\|_F^2 &= \|U_k Z^{-1} U_k^T S^T S A - A\|_F^2 \\ &= \|U_k U_k^T (U_k Z^{-1} U_k^T S^T S A - A)\|_F^2 + \|(I - U_k U_k^T)(U_k Z^{-1} U_k^T S^T S A - A)\|_F^2 \\ &= \|U_k Z^{-1} U_k^T S^T S A - A_k\|_F^2 + \|A - A_k\|_F^2 \quad (\text{why?}) \end{aligned}$$

Pythagorean
identity with
 $U_k U_k^T$



LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|BSA - A\|_F^2 \leq (1 + O(k^2 \varepsilon)) \|A - A_k\|_F^2$

- We have:

Pythagorean identity with $U_k U_k^T$ 

$$\begin{aligned} \|BSA - A\|_F^2 &= \|U_k Z^{-1} U_k^T S^T S A - A\|_F^2 \\ &= \|U_k U_k^T (U_k Z^{-1} U_k^T S^T S A - A)\|_F^2 + \|(I - U_k U_k^T)(U_k Z^{-1} U_k^T S^T S A - A)\|_F^2 \\ &= \|U_k Z^{-1} U_k^T S^T S A - A_k\|_F^2 + \|A - A_k\|_F^2 \quad (\text{why?}) \end{aligned}$$

- Therefore, we need to show: $\|U_k Z^{-1} U_k^T S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$
- We have:
 - $\|U_k Z^{-1} U_k^T S^T S A - A_k\|_F^2 =$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$
- We have:
 - $\|U_k Z^{-1} U_k^T S^T S A - A_k\|_F^2 = \|U_k Z^{-1} U_k S^T S (A_k + A - A_k) - A_k\|_F^2$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$
- We have:
 - $\|U_k Z^{-1} U_k^T S^T S A - A_k\|_F^2 = \|U_k Z^{-1} U_k S^T S (A_k + A - A_k) - A_k\|_F^2$
 - $= \|U_k Z^{-1} U_k S^T S A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$
- We have:
 - $\|U_k Z^{-1} U_k^T S^T S A - A_k\|_F^2 = \|U_k Z^{-1} U_k S^T S (A_k + A - A_k) - A_k\|_F^2$
 - $= \|U_k Z^{-1} U_k S^T S A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2$
 - $= \|U_k Z^{-1} U_k S^T S U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$

- We have:

- $\|U_k Z^{-1} U_k^T S^T S A - A_k\|_F^2 = \|U_k Z^{-1} U_k S^T S (A_k + A - A_k) - A_k\|_F^2$
- $= \|U_k Z^{-1} U_k S^T S A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2$
- $= \|U_k Z^{-1} U_k S^T S U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2$
- $= \|U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2$

$$Z^{-1} U_k S^T S U_k = I$$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$

- We have:

$$\begin{aligned} & \bullet \left\| U_k Z^{-1} U_k^T S^T S A - A_k \right\|_F^2 = \left\| U_k Z^{-1} U_k S^T S (A_k + A - A_k) - A_k \right\|_F^2 \\ & \bullet = \left\| U_k Z^{-1} U_k S^T S A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k \right\|_F^2 \\ & \bullet = \left\| U_k Z^{-1} U_k S^T S U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k \right\|_F^2 \\ & \bullet = \left\| U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k \right\|_F^2 \\ & \bullet = \left\| A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k \right\|_F^2 \end{aligned}$$

$$Z^{-1} U_k S^T S U_k = I$$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$

- We have:

$$\begin{aligned} & \bullet \left\| U_k Z^{-1} U_k^T S^T S A - A_k \right\|_F^2 = \left\| U_k Z^{-1} U_k S^T S (A_k + A - A_k) - A_k \right\|_F^2 \\ & \bullet = \left\| U_k Z^{-1} U_k S^T S A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k \right\|_F^2 \\ & \bullet = \left\| U_k Z^{-1} U_k S^T S U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k \right\|_F^2 \\ & \bullet = \left\| U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k \right\|_F^2 \\ & \bullet = \left\| \textcolor{blue}{A}_k + U_k Z^{-1} U_k S^T S (A - A_k) - \textcolor{blue}{A}_k \right\|_F^2 \\ & \bullet = \left\| U_k Z^{-1} U_k S^T S (A - A_k) \right\|_F^2 \end{aligned}$$

$$Z^{-1} U_k S^T S U_k = I$$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$

- We have:

$$\begin{aligned} \bullet \quad & \|U_k Z^{-1} U_k^T S^T S A - A_k\|_F^2 = \|U_k Z^{-1} U_k S^T S (A_k + A - A_k) - A_k\|_F^2 \\ \bullet \quad & = \|U_k Z^{-1} U_k S^T S A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2 \\ \bullet \quad & = \|U_k Z^{-1} U_k S^T S U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2 \\ \bullet \quad & = \|U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2 \\ \bullet \quad & = \|A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2 \\ & = \|U_k Z^{-1} U_k S^T S (A - A_k)\|_F^2 \end{aligned}$$

$$Z^{-1} U_k S^T S U_k = I$$

- Therefore, we need to show: $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} U_k^T S^T (A - A_k)\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$
- We have:
 - $\|U_k Z^{-1} U_k^T S^T S (A - A_k)\|_F^2 \leq \|U_k\|_2^2 \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S (A - A_k)\|_F^2$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$
- We have:
 - $\|U_k Z^{-1} U_k S^T S (A - A_k)\|_F^2 \leq \|U_k\|_2^2 \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S (A - A_k)\|_F^2$
 - $= k \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S (A - A_k)\|_F^2$
- (Why $\|U_k\|_2^2 = k$?) 

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} U_k^T S^T (A - A_k)\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$
- We have:
 - $\|U_k Z^{-1} U_k^T S^T S (A - A_k)\|_F^2 \leq \|U_k\|_2^2 \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S (A - A_k)\|_F^2$
 - $= k \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S (A - A_k)\|_F^2$
 - (Why $\|U_k\|_2^2 = k$?) 
- To finish the proof, we will bound both $\|Z^{-1}\|_2^2$ and $\|U_k^T S^T S (A - A_k)\|_F^2$ using AMM

LRA Accuracy: Proof

- We need to bound $\|U_k^T S^T S(A - A_k)\|_F^2$

LRA Accuracy: Proof

- We need to bound $\|U_k^T S^T S(A - A_k)\|_F^2$
- Recall we assume: S satisfies **AMM**

LRA Accuracy: Proof

- We need to bound $\|U_k^T S^T S(A - A_k)\|_F^2$
- Recall we assume: S satisfies **AMM**
- Therefore: $\|U_k^T S^T S(A - A_k) - U_k(A - A_k)\|_F^2 \leq \varepsilon \|U_k\|_F^2 \|A - A_k\|_F^2 = \varepsilon k \|A - A_k\|_F^2$

LRA Accuracy: Proof

- We need to bound $\|U_k^T S^T S(A - A_k)\|_F^2$
- Recall we assume: S satisfies **AMM**
- Therefore: $\|U_k^T S^T S(A - A_k) - U_k(A - A_k)\|_F^2 \leq \varepsilon \|U_k\|_F^2 \|A - A_k\|_F^2 = \varepsilon k \|A - A_k\|_F^2$
- Observe: $U_k(A - A_k) = 0$ (why?)

LRA Accuracy: Proof

- We need to bound $\|U_k^T S^T S(A - A_k)\|_F^2$
- Recall we assume: S satisfies **AMM**
- Therefore: $\|U_k^T S^T S(A - A_k) - U_k(A - A_k)\|_F^2 \leq \varepsilon \|U_k\|_F^2 \|A - A_k\|_F^2 = \varepsilon k \|A - A_k\|_F^2$
- Observe: $U_k(A - A_k) = 0$ (why?)
- Therefore: $\|U_k^T S^T S(A - A_k)\|_F^2 \leq \varepsilon k \|A - A_k\|_F^2$

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- Recall we assume: S satisfies [AMM](#)

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- Recall we assume: S satisfies [AMM](#)
- Therefore: $\|U_k^T S^T S U_k - U_k^T U_k\|_F^2 \leq \varepsilon \|U_k\|_F^4$

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- Recall we assume: S satisfies [AMM](#)
- Therefore: $\|U_k^T S^T S U_k - U_k^T U_k\|_F^2 \leq \varepsilon \|U_k\|_F^4 = ???$

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- Recall we assume: S satisfies [AMM](#)
- Therefore: $\|U_k^T S^T S U_k - U_k^T U_k\|_F^2 \leq \varepsilon \|U_k\|_F^4 = \varepsilon k^2$

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- Recall we assume: S satisfies [AMM](#)
- Therefore: $\|U_k^T S^T S U_k - U_k^T U_k\|_F^2 \leq \varepsilon \|U_k\|_F^4 = \varepsilon k^2$
- Observe: $U_k^T U_k = I$ (why?)

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- Recall we assume: S satisfies [AMM](#)
- Therefore: $\|U_k^T S^T S U_k - U_k^T U_k\|_F^2 \leq \varepsilon \|U_k\|_F^4 = \varepsilon k^2$
- Observe: $U_k^T U_k = I$ (why?)
- Therefore: $\|U_k^T S^T S U_k - I\|_F^2 \leq \varepsilon k^2$

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- Recall we assume: S satisfies [AMM](#)
- Therefore: $\|U_k^T S^T S U_k - U_k^T U_k\|_F^2 \leq \varepsilon \|U_k\|_F^4 = \varepsilon k^2$
- Observe: $U_k^T U_k = I$ (why?)
- Therefore: $\|U_k^T S^T S U_k - I\|_F^2 \leq \varepsilon k^2$
- Therefore: $\|U_k^T S^T S U_k - I\|_2^2 \leq \varepsilon k^2$ (why?) (note we moved from Frobenius to spectral norm)

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- Recall we assume: S satisfies [AMM](#)
- Therefore: $\|U_k^T S^T S U_k - U_k^T U_k\|_F^2 \leq \varepsilon \|U_k\|_F^4 = \varepsilon k^2$
- Observe: $U_k^T U_k = I$ (why?)
- Therefore: $\|U_k^T S^T S U_k - I\|_F^2 \leq \varepsilon k^2$
- Therefore: $\|U_k^T S^T S U_k - I\|_2^2 \leq \varepsilon k^2$ (why?) (note we moved from Frobenius to spectral norm)
- We choose ε small enough so that $\|U_k^T S^T S U_k - I\|_2^2 \leq 1/9$

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- We showed: $\|U_k^T S^T S U_k - I\|_2^2 \leq 1/9$

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- We showed: $\|U_k^T S^T S U_k - I\|_2^2 \leq 1/9$
- **Lemma:** If a square matrix M satisfies $\|M - I\|_2 \leq \alpha$ for $\alpha < 1$, then all of the eigenvalues of M are in $[1 - \alpha, 1 + \alpha]$. (Proof?)

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- We showed: $\|U_k^T S^T S U_k - I\|_2^2 \leq 1/9$
- **Lemma:** If a square matrix M satisfies $\|M - I\|_2 \leq \alpha$ for $\alpha < 1$, then all of the eigenvalues of M are in $[1 - \alpha, 1 + \alpha]$. (Proof?)
- **Corollary:** All eigenvalues of Z are between $2/3$ and $4/3$.

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- We showed: $\|U_k^T S^T S U_k - I\|_2^2 \leq 1/9$
- **Lemma:** If a square matrix M satisfies $\|M - I\|_2 \leq \alpha$ for $\alpha < 1$, then all of the eigenvalues of M are in $[1 - \alpha, 1 + \alpha]$. (Proof?)
- **Corollary:** All eigenvalues of Z are between $2/3$ and $4/3$.
- In particular:
 - Z is invertible.
 - $\|Z^{-1}\|_2^2 \leq \frac{16}{9} = O(1)$.

LRA Accuracy: End of Proof (Finally!)

LRA Accuracy: End of Proof (Finally!)

- Our goal was to prove: $\|BSA - A\|_F^2 \leq (1 + O(k^2\varepsilon))\|A - A_k\|_F^2$

LRA Accuracy: End of Proof (Finally!)

- Our goal was to prove: $\|BSA - A\|_F^2 \leq (1 + O(k^2\varepsilon))\|A - A_k\|_F^2$
- We showed it's enough to prove: $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2\varepsilon)\|A - A_k\|_F^2$

LRA Accuracy: End of Proof (Finally!)

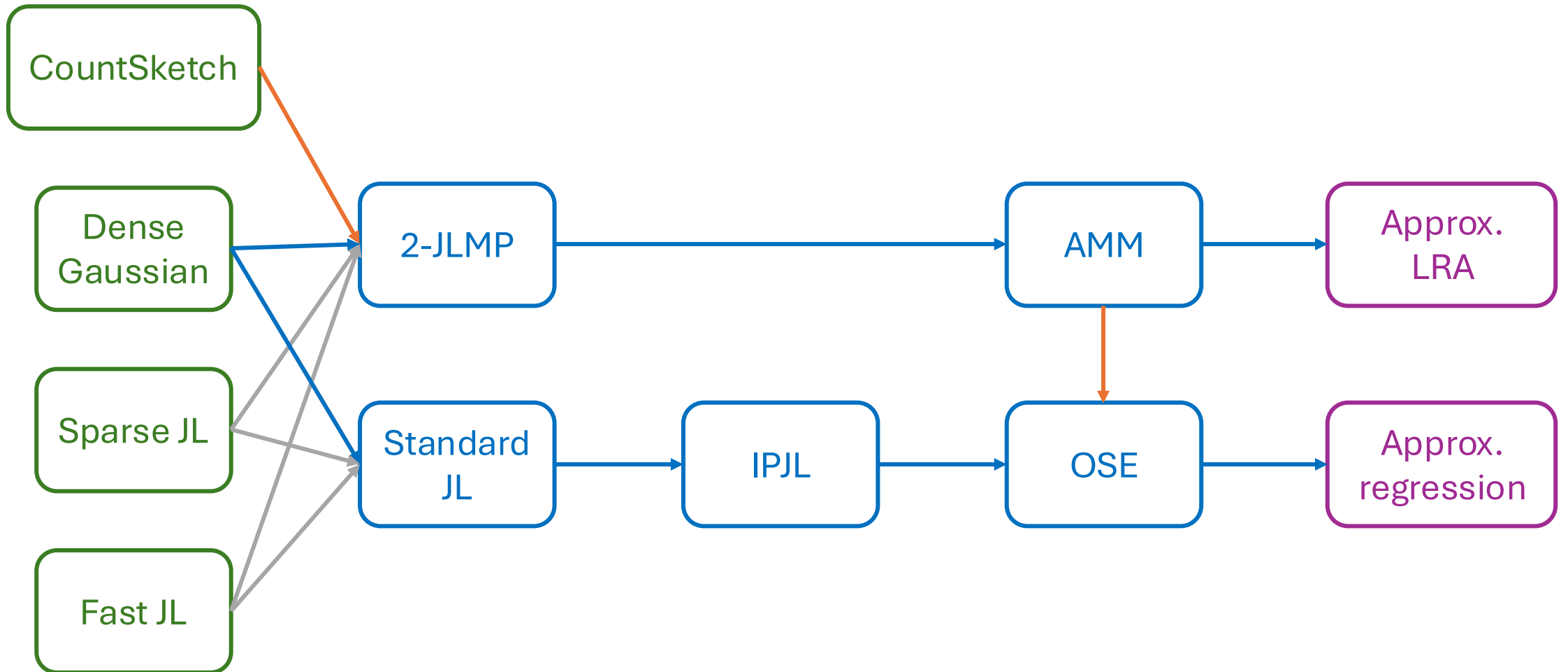
- Our goal was to prove: $\|BSA - A\|_F^2 \leq (1 + O(k^2\varepsilon))\|A - A_k\|_F^2$
- We showed it's enough to prove: $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2\varepsilon)\|A - A_k\|_F^2$
- We proved:
 - $\|U_k Z^{-1} U_k S^T S(A - A_k)\|_F^2 \leq k \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S(A - A_k)\|_F^2$
 - $\|U_k^T S^T S(A - A_k)\|_F^2 \leq \varepsilon k \|A - A_k\|_F^2$
 - $\|Z^{-1}\|_2^2 = O(1)$

LRA Accuracy: End of Proof (Finally!)

- Our goal was to prove: $\|BSA - A\|_F^2 \leq (1 + O(k^2\varepsilon))\|A - A_k\|_F^2$
- We showed it's enough to prove: $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2\varepsilon)\|A - A_k\|_F^2$
- We proved:
 - $\|U_k Z^{-1} U_k S^T S(A - A_k)\|_F^2 \leq k \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S(A - A_k)\|_F^2$
 - $\|U_k^T S^T S(A - A_k)\|_F^2 \leq \varepsilon k \|A - A_k\|_F^2$
 - $\|Z^{-1}\|_2^2 = O(1)$
- Which together yields: $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2\varepsilon)\|A - A_k\|_F^2$ as needed.



Efficient Linear Algebra: Summary



Efficient Linear Algebra: Summary

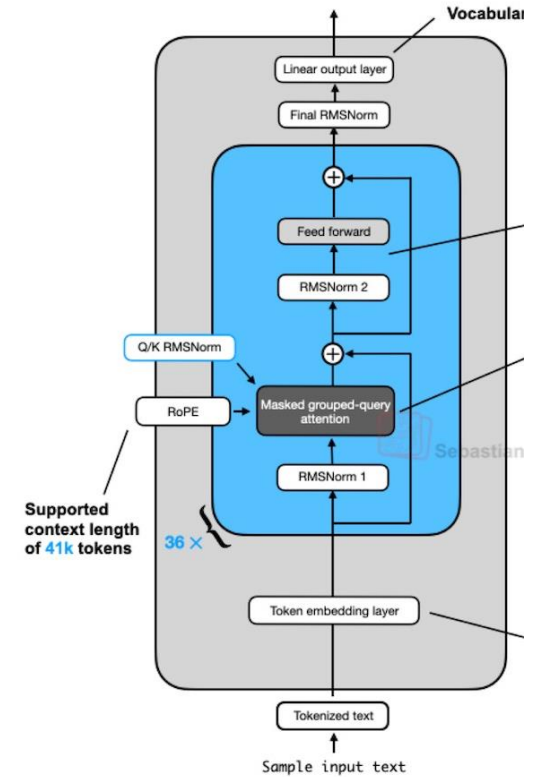
- JL and JL-like techniques are useful for randomized approximate algorithms for linear algebra problems
 - Matrix multiplication
 - Regression
 - Low rank approximation

LoRA: Low Rank Adaptation in LLMs

An Aside

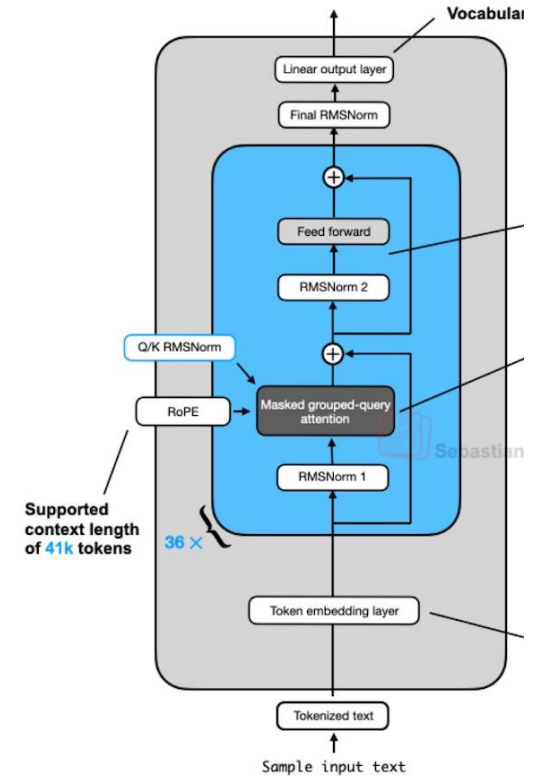
Recap: Attention in LLMs

- Attention:
 - Input: $K, Q, V \in \mathbb{R}^{n \times d}$
 - Output: $\text{softmax}\left(\frac{1}{\sqrt{d}} QK^T\right) V$



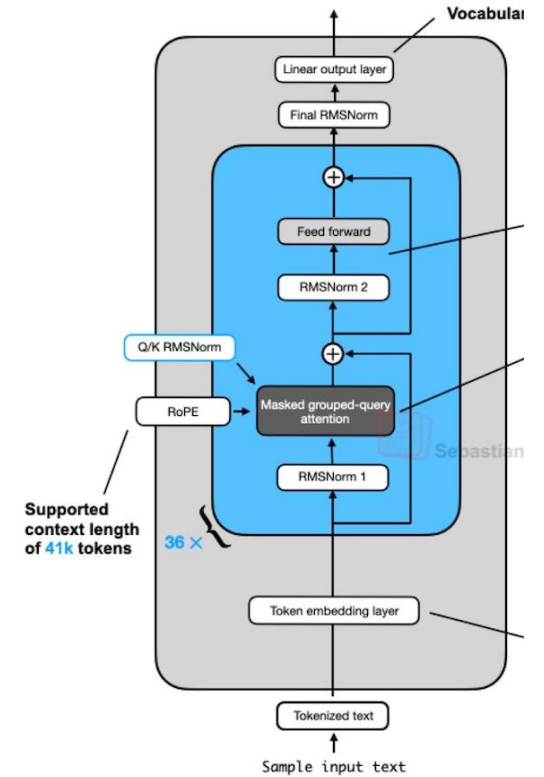
Recap: Attention in LLMs

- Attention:
 - Input: $K, Q, V \in \mathbb{R}^{n \times d}$
 - Output: $\text{softmax}\left(\frac{1}{\sqrt{d}} QK^T\right) V$
- Where do $K, Q, V \in \mathbb{R}^{n \times d}$ come from?



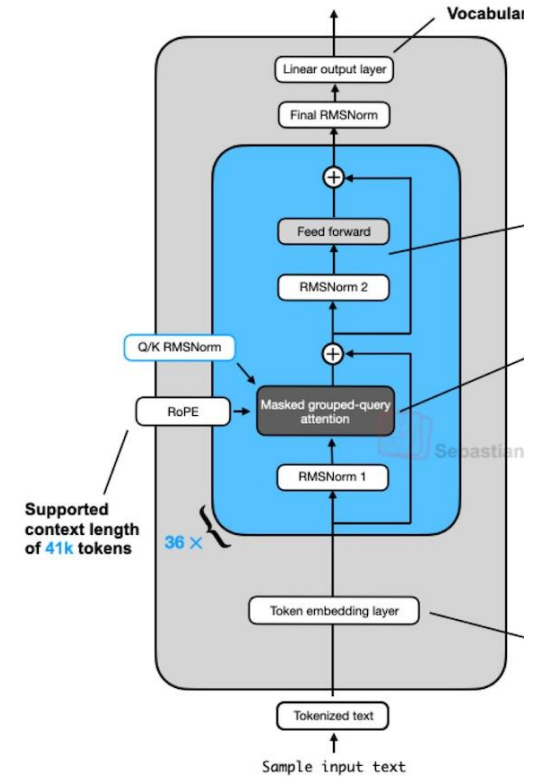
Recap: Attention in LLMs

- Attention:
 - Input: $K, Q, V \in \mathbb{R}^{n \times d}$
 - Output: $\text{softmax}\left(\frac{1}{\sqrt{d}} QK^T\right) V$
- Where do $K, Q, V \in \mathbb{R}^{n \times d}$ come from?
- Focusing on self-attention:



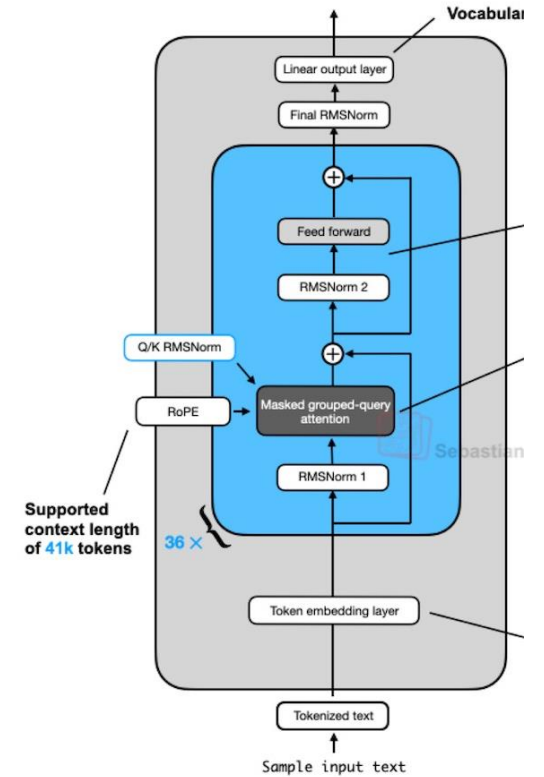
Recap: Attention in LLMs

- Attention:
 - Input: $K, Q, V \in \mathbb{R}^{n \times d}$
 - Output: $\text{softmax}\left(\frac{1}{\sqrt{d}} QK^T\right) V$
- Where do $K, Q, V \in \mathbb{R}^{n \times d}$ come from?
- Focusing on self-attention:
 - The input is $X \in \mathbb{R}^{n \times d'}$, representing d' -dimensional embeddings of n tokens



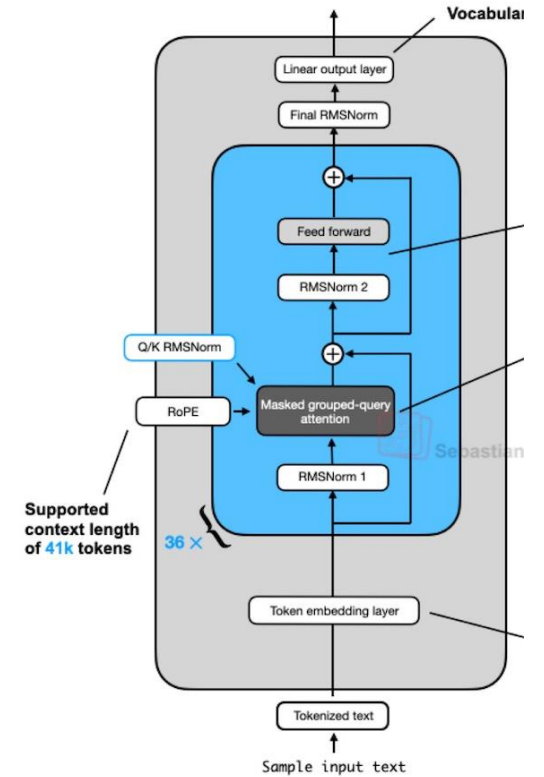
Recap: Attention in LLMs

- Attention:
 - Input: $K, Q, V \in \mathbb{R}^{n \times d}$
 - Output: $\text{softmax}\left(\frac{1}{\sqrt{d}} QK^T\right) V$
- Where do $K, Q, V \in \mathbb{R}^{n \times d}$ come from?
- Focusing on self-attention:
 - The input is $X \in \mathbb{R}^{n \times d'}$, representing d' -dimensional embeddings of n tokens
 - Attention stores three internal parameter matrices: $W_K, W_Q, W_V \in \mathbb{R}^{d' \times d}$
 - They are learned at training and hard-coded at inference



Recap: Attention in LLMs

- Attention:
 - Input: $K, Q, V \in \mathbb{R}^{n \times d}$
 - Output: $\text{softmax}\left(\frac{1}{\sqrt{d}} QK^T\right) V$
- Where do $K, Q, V \in \mathbb{R}^{n \times d}$ come from?
- Focusing on self-attention:
 - The input is $X \in \mathbb{R}^{n \times d'}$, representing d' -dimensional embeddings of n tokens
 - Attention stores three internal parameter matrices: $W_K, W_Q, W_V \in \mathbb{R}^{d' \times d}$
 - They are learned at training and hard-coded at inference
 - Upon getting X it computes: $K = XW_K, Q = XW_Q, V = XW_V$



LLM Training Pipeline

LLM Training Pipeline

- Traditional ML:

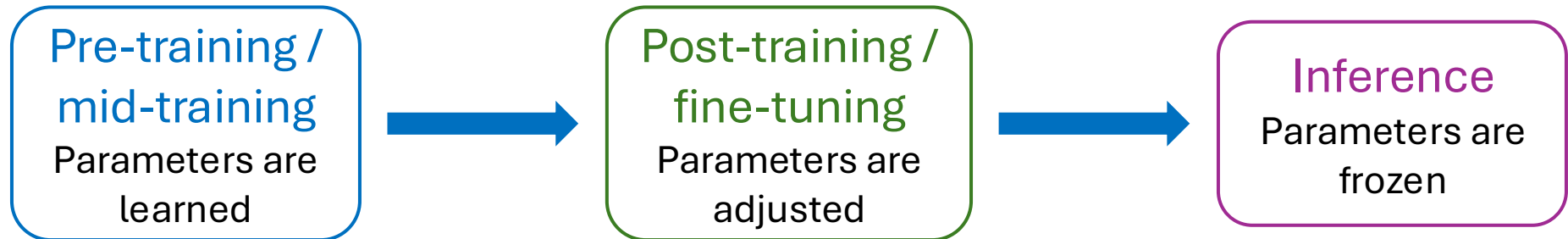


LLM Training Pipeline

- Traditional ML:



- LLMs:

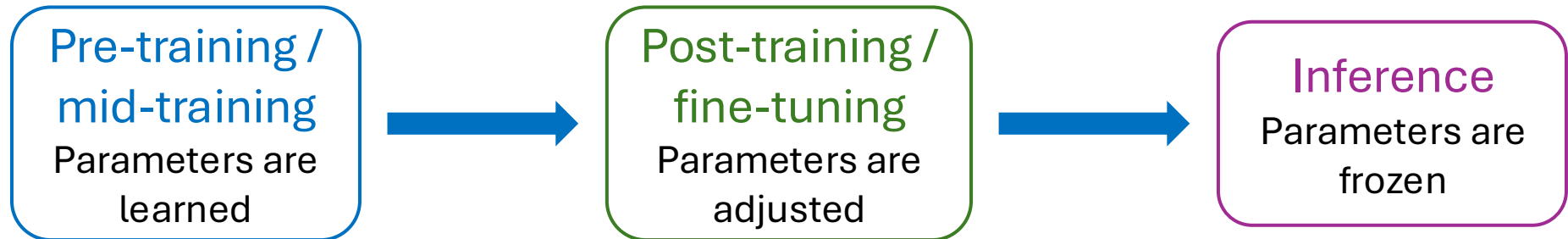


LLM Training Pipeline

- Traditional ML:



- LLMs:



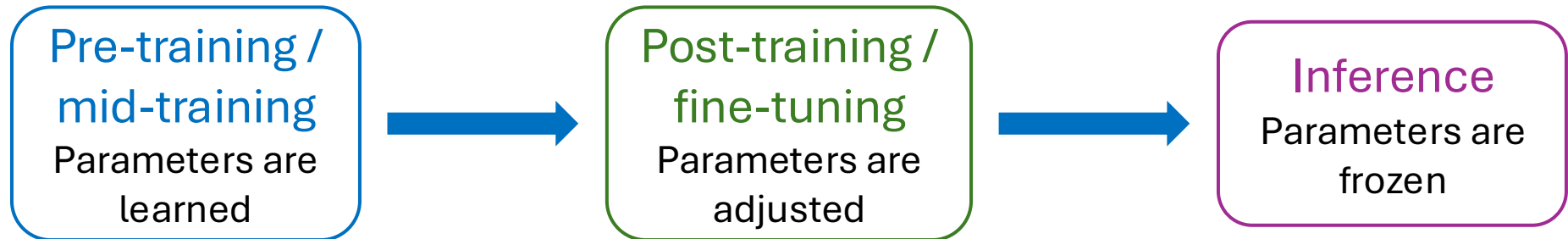
- During post-training / fine-tuning, we want to adjust a small number of parameters

LLM Training Pipeline

- Traditional ML:

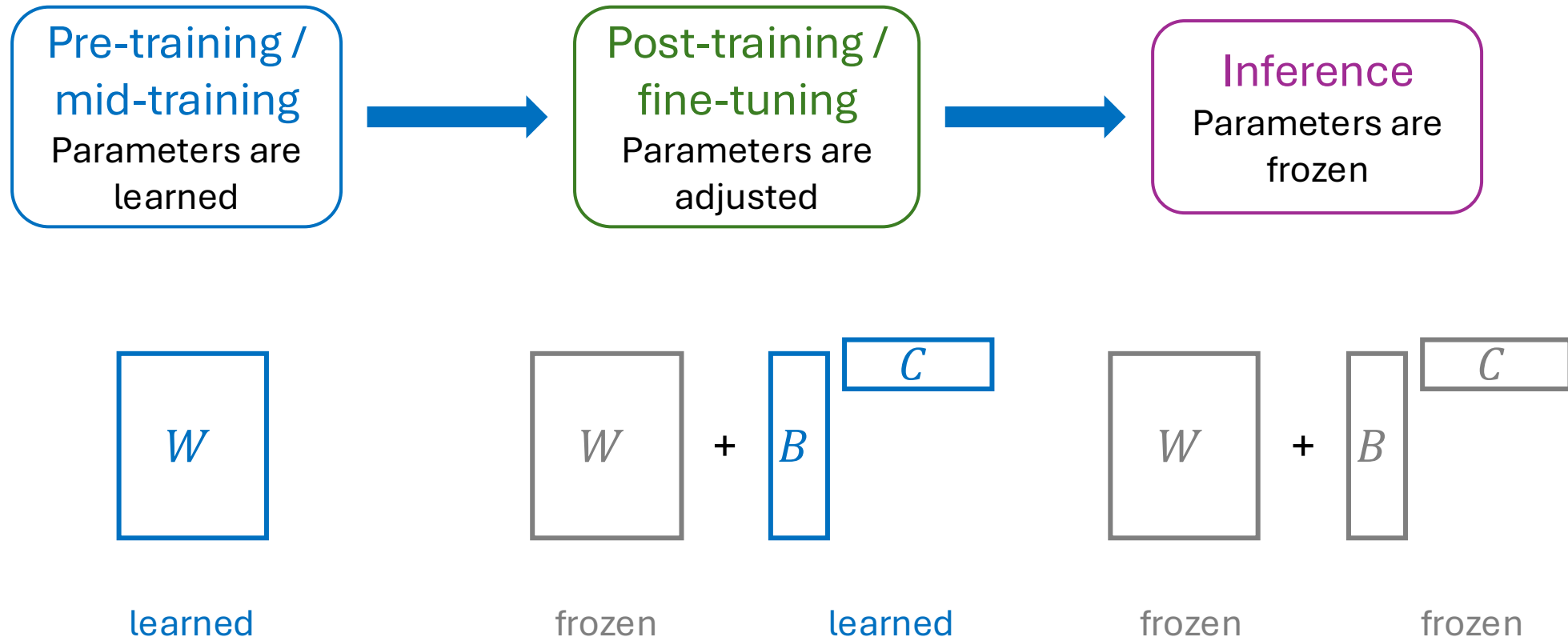


- LLMs:



- During post-training / fine-tuning, we want to adjust a small number of parameters
 - Why?

LoRA: Low Rank Adaptation

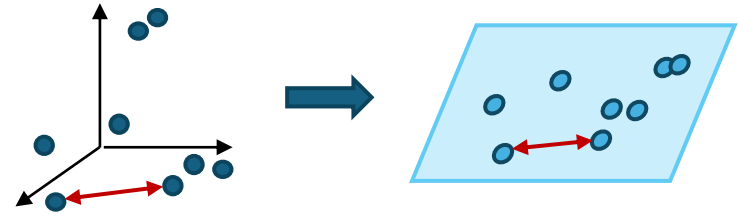


Questions?

Fast Johnson-Lindenstrass Transform (FJLT)

Dimension reduction in nearly linear time

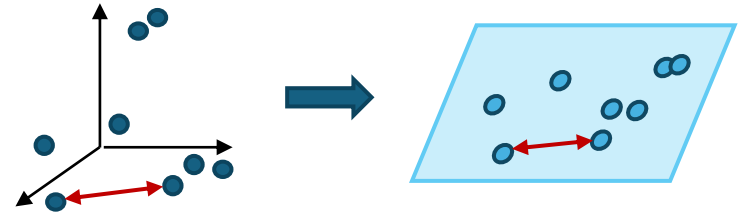
Reminder: JL and DJL



- **Lemma (DJL):** Let $\varepsilon, \delta \in (0,1)$ and $m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$. Let $M \in \mathbb{R}^{m \times d}$ be a matrix with iid entries from $N(0,1)$. For every unit vector $x \in \mathbb{R}^d$,

$$\Pr \left[1 - \varepsilon \leq \left\| \frac{1}{\sqrt{m}} Mx \right\|_2 \leq 1 + \varepsilon \right] \geq 1 - \delta$$

Reminder: JL and DJL

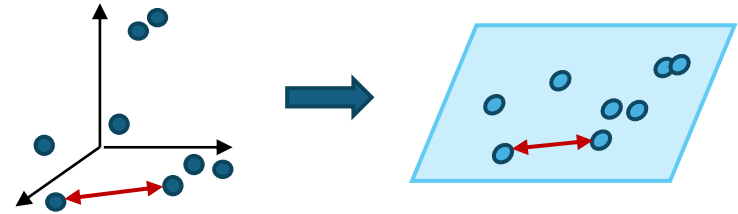


- **Lemma (DJL):** Let $\varepsilon, \delta \in (0,1)$ and $m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$. Let $M \in \mathbb{R}^{m \times d}$ be a matrix with iid entries from $N(0,1)$. For every unit vector $x \in \mathbb{R}^d$,

$$\Pr \left[1 - \varepsilon \leq \left\| \frac{1}{\sqrt{m}} Mx \right\|_2 \leq 1 + \varepsilon \right] \geq 1 - \delta$$

- **Recall:** DJL $\Rightarrow$ JL

Reminder: JL and DJL

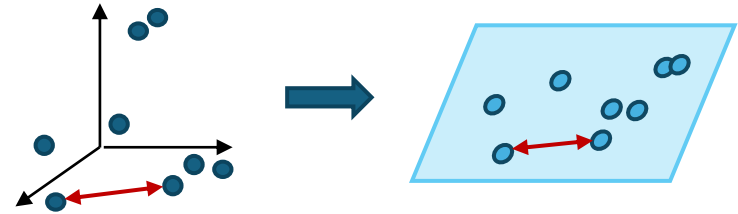


- **Lemma (DJL):** Let $\varepsilon, \delta \in (0,1)$ and $m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$. Let $M \in \mathbb{R}^{m \times d}$ be a matrix with iid entries from $N(0,1)$. For every unit vector $x \in \mathbb{R}^d$,

$$\Pr \left[1 - \varepsilon \leq \left\| \frac{1}{\sqrt{m}} Mx \right\|_2 \leq 1 + \varepsilon \right] \geq 1 - \delta$$

- **Recall:** DJL $\Rightarrow$ JL
- Time to reduce the dimension of a given $x \in \mathbb{R}^d$: ???

Reminder: JL and DJL

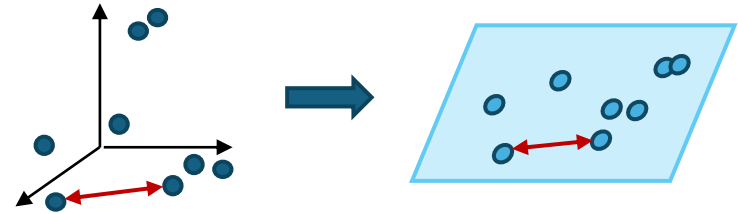


- **Lemma (DJL):** Let $\varepsilon, \delta \in (0,1)$ and $m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$. Let $M \in \mathbb{R}^{m \times d}$ be a matrix with iid entries from $N(0,1)$. For every unit vector $x \in \mathbb{R}^d$,

$$\Pr \left[1 - \varepsilon \leq \left\| \frac{1}{\sqrt{m}} Mx \right\|_2 \leq 1 + \varepsilon \right] \geq 1 - \delta$$

- **Recall:** DJL $\Rightarrow$ JL
- Time to reduce the dimension of a given $x \in \mathbb{R}^d$: $O(dm)$

Reminder: JL and DJL

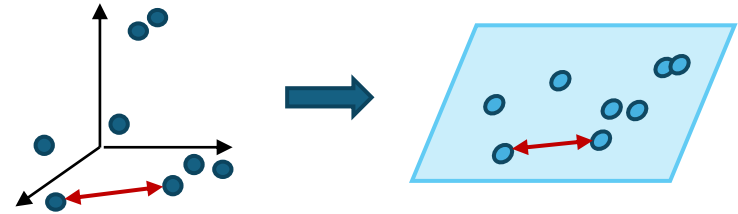


- **Lemma (DJL):** Let $\varepsilon, \delta \in (0,1)$ and $m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$. Let $M \in \mathbb{R}^{m \times d}$ be a matrix with iid entries from $N(0,1)$. For every unit vector $x \in \mathbb{R}^d$,

$$\Pr \left[1 - \varepsilon \leq \left\| \frac{1}{\sqrt{m}} Mx \right\|_2 \leq 1 + \varepsilon \right] \geq 1 - \delta$$

- **Recall:** DJL $\Rightarrow$ JL
- Time to reduce the dimension of a given $x \in \mathbb{R}^d$: $O(dm)$
- **Fast JL** [Ailon & Chazelle '06]: time $\tilde{O}(d + m)$ with almost the same m : $m = \tilde{O}(1/\varepsilon^2)$

Reminder: JL and DJL



- **Lemma (DJL):** Let $\varepsilon, \delta \in (0,1)$ and $m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$. Let $M \in \mathbb{R}^{m \times d}$ be a matrix with iid entries from $N(0,1)$. For every unit vector $x \in \mathbb{R}^d$,

$$\Pr \left[1 - \varepsilon \leq \left\| \frac{1}{\sqrt{m}} Mx \right\|_2 \leq 1 + \varepsilon \right] \geq 1 - \delta$$

- **Recall:** DJL $\Rightarrow$ JL
- Time to reduce the dimension of a given $x \in \mathbb{R}^d$: $O(dm)$
- **Fast JL** [Ailon & Chazelle '06]: time $\tilde{O}(d + m)$ with almost the same m : $m = \tilde{O}(1/\varepsilon^2)$
- We will see:
 - Target dimension: $m = O\left(\frac{\log^2(d/\delta) \log(1/\delta)}{\varepsilon^2}\right)$
 - Time per point: $O(d \log d + m)$

Intuition (1): Fast Random Projection?

Intuition (1): Fast Random Projection?

- Goal: Compute Mx for a random matrix $M \in \mathbb{R}^{m \times d}$ and an arbitrary vector $x \in \mathbb{R}^d$
 - M should intuitively simulate a random projection

Intuition (1): Fast Random Projection?

- Goal: Compute Mx for a random matrix $M \in \mathbb{R}^{m \times d}$ and an arbitrary vector $x \in \mathbb{R}^d$
 - M should intuitively simulate a random projection
- How can we make sure computing Mx is faster than the naïve time $O(nd)$?

Intuition (1): Fast Random Projection?

- Goal: Compute Mx for a random matrix $M \in \mathbb{R}^{m \times d}$ and an arbitrary vector $x \in \mathbb{R}^d$
 - M should intuitively simulate a random projection
- How can we make sure computing Mx is faster than the naïve time $O(nd)$?
- We require M to have a special structure that enables fast multiplication by a vector

Intuition (1): Fast Random Projection?

- Goal: Compute Mx for a random matrix $M \in \mathbb{R}^{m \times d}$ and an arbitrary vector $x \in \mathbb{R}^d$
 - M should intuitively simulate a random projection
- How can we make sure computing Mx is faster than the naïve time $O(nd)$?
- We require M to have a special structure that enables fast multiplication by a vector
- Examples?

Intuition (1): Fast Random Projection?

- Goal: Compute Mx for a random matrix $M \in \mathbb{R}^{m \times d}$ and an arbitrary vector $x \in \mathbb{R}^d$
 - M should intuitively simulate a random projection
- How can we make sure computing Mx is faster than the naïve time $O(nd)$?
- We require M to have a special structure that enables fast multiplication by a vector
- Examples?
 - Sparsity: e.g., what if M is diagonal?

Intuition (1): Fast Random Projection?

- Goal: Compute Mx for a random matrix $M \in \mathbb{R}^{m \times d}$ and an arbitrary vector $x \in \mathbb{R}^d$
 - M should intuitively simulate a random projection
- How can we make sure computing Mx is faster than the naïve time $O(nd)$?
- We require M to have a special structure that enables fast multiplication by a vector
- Examples?
 - Sparsity: e.g., what if M is diagonal?
 - Non-sparse structure: e.g., what if all rows of M are the same?

Intuition (1): Fast Random Projection?

- Goal: Compute Mx for a random matrix $M \in \mathbb{R}^{m \times d}$ and an arbitrary vector $x \in \mathbb{R}^d$
 - M should intuitively simulate a random projection
- How can we make sure computing Mx is faster than the naïve time $O(nd)$?
- We require M to have a special structure that enables fast multiplication by a vector
- Examples?
 - Sparsity: e.g., what if M is diagonal?
 - Non-sparse structure: e.g., what if all rows of M are the same?
- But we also need M to “simulate a random projection”

Intuition (2): Flattening Coordinates

Intuition (2): Flattening Coordinates

- Idea for fast JL: Let $x \in \mathbb{R}^d$

Intuition (2): Flattening Coordinates

- Idea for fast JL: Let $x \in \mathbb{R}^d$
 - We want to preserve the (squared) Euclidean norm: $\|x\|_2^2$

Intuition (2): Flattening Coordinates

- Idea for fast JL: Let $x \in \mathbb{R}^d$
 - We want to preserve the (squared) Euclidean norm: $\|x\|_2^2$
 - It decomposes additively: $\|x\|_2^2 = \sum_{i=1}^d x_i^2$

Intuition (2): Flattening Coordinates

- Idea for fast JL: Let $x \in \mathbb{R}^d$
 - We want to preserve the (squared) Euclidean norm: $\|x\|_2^2$
 - It decomposes additively: $\|x\|_2^2 = \sum_{i=1}^d x_i^2$
 - We can estimate by subsampling coordinates:
 - $X = d \cdot x_i^2$ where $i \sim \text{uniform}\{1, \dots, d\}$ is an unbiased estimator

Intuition (2): Flattening Coordinates

- Idea for fast JL: Let $x \in \mathbb{R}^d$
 - We want to preserve the (squared) Euclidean norm: $\|x\|_2^2$
 - It decomposes additively: $\|x\|_2^2 = \sum_{i=1}^d x_i^2$
 - We can estimate by subsampling coordinates:
 - $X = d \cdot x_i^2$ where $i \sim \text{uniform}\{1, \dots, d\}$ is an unbiased estimator
 - The variance of this estimator is smaller when the x_i 's are as small as possible

Intuition (2): Flattening Coordinates

- Idea for fast JL: Let $x \in \mathbb{R}^d$
 - We want to preserve the (squared) Euclidean norm: $\|x\|_2^2$
 - It decomposes additively: $\|x\|_2^2 = \sum_{i=1}^d x_i^2$
 - We can estimate by subsampling coordinates:
 - $X = d \cdot x_i^2$ where $i \sim \text{uniform}\{1, \dots, d\}$ is an unbiased estimator
 - The variance of this estimator is smaller when the x_i 's are as small as possible
 - How small can they be?

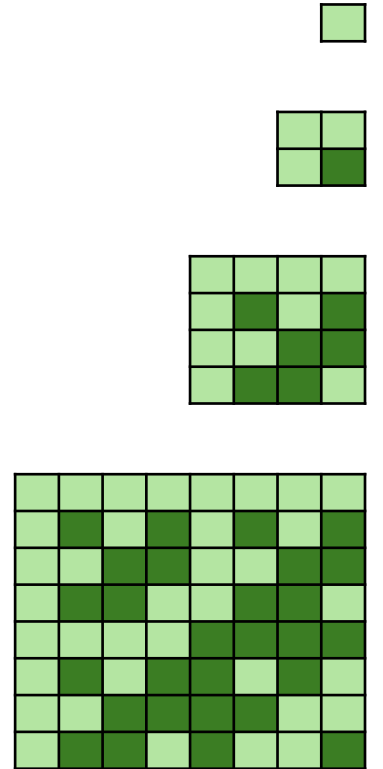
Intuition (2): Flattening Coordinates

- Idea for fast JL: Let $x \in \mathbb{R}^d$
 - We want to preserve the (squared) Euclidean norm: $\|x\|_2^2$
 - It decomposes additively: $\|x\|_2^2 = \sum_{i=1}^d x_i^2$
 - We can estimate by subsampling coordinates:
 - $X = d \cdot x_i^2$ where $i \sim \text{uniform}\{1, \dots, d\}$ is an unbiased estimator
 - The variance of this estimator is smaller when the x_i 's are as small as possible
 - How small can they be?
- **Upshot:** if we could flatten the coordinates of x , to as close to $\|x\|_2^2 / \sqrt{d}$ as possible, we could uniformly sample few coordinates for dimension reduction

The Hadamard Matrix

- The Hadamard matrix of order $2^\ell \times 2^\ell$ is defined by induction:

$$H_{2^0} = [1], \quad H_{2^\ell} = \frac{1}{\sqrt{2}} \begin{bmatrix} H_{2^{\ell-1}} & H_{2^{\ell-1}} \\ H_{2^{\ell-1}} & -H_{2^{\ell-1}} \end{bmatrix}$$

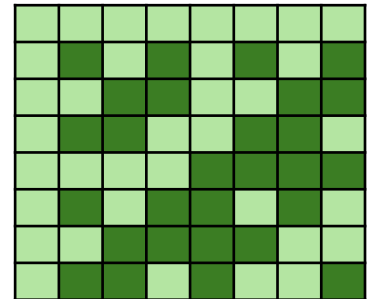
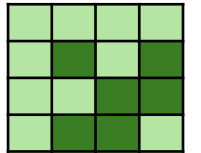


The Hadamard Matrix

- The **Hadamard** matrix of order $2^\ell \times 2^\ell$ is defined by induction:

$$H_{2^0} = [1], \quad H_{2^\ell} = \frac{1}{\sqrt{2}} \begin{bmatrix} H_{2^{\ell-1}} & H_{2^{\ell-1}} \\ H_{2^{\ell-1}} & -H_{2^{\ell-1}} \end{bmatrix}$$

- We can assume w.l.o.g. that d is a power of 2 (why?)

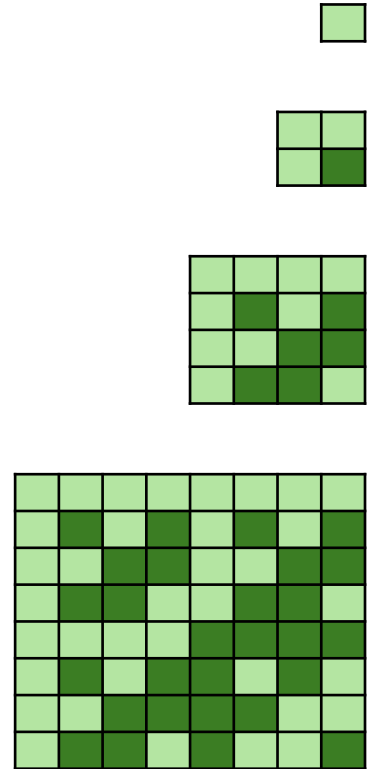


The Hadamard Matrix

- The **Hadamard** matrix of order $2^\ell \times 2^\ell$ is defined by induction:

$$H_{2^0} = [1], \quad H_{2^\ell} = \frac{1}{\sqrt{2}} \begin{bmatrix} H_{2^{\ell-1}} & H_{2^{\ell-1}} \\ H_{2^{\ell-1}} & -H_{2^{\ell-1}} \end{bmatrix}$$

- We can assume w.l.o.g. that d is a power of 2 (**why?**)
- Henceforce we will denote $H := H_d$

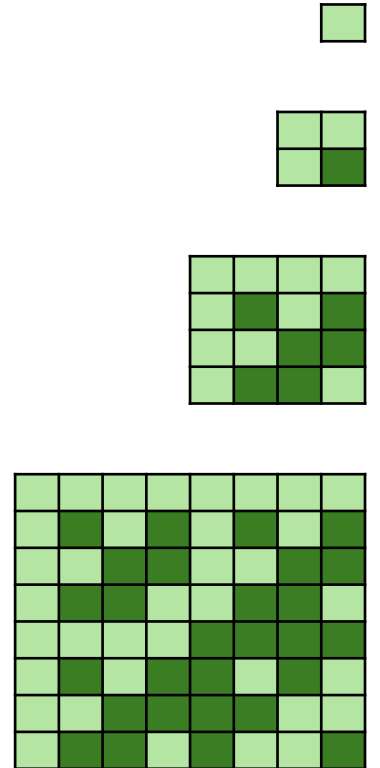


The Hadamard Matrix

- The **Hadamard** matrix of order $2^\ell \times 2^\ell$ is defined by induction:

$$H_{2^0} = [1], \quad H_{2^\ell} = \frac{1}{\sqrt{2}} \begin{bmatrix} H_{2^{\ell-1}} & H_{2^{\ell-1}} \\ H_{2^{\ell-1}} & -H_{2^{\ell-1}} \end{bmatrix}$$

- We can assume w.l.o.g. that d is a power of 2 (**why?**)
- Henceforce we will denote $H := H_d$
- Claims:
 - H is an **orthogonal matrix**: $H^T H = H H^T = I$ (**proof?**)



The Hadamard Matrix

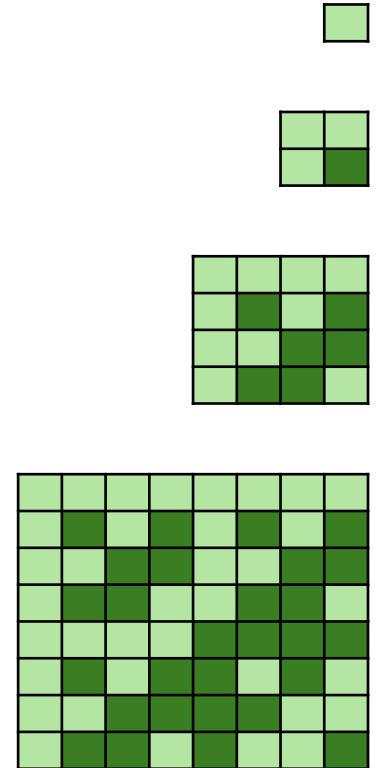
- The **Hadamard** matrix of order $2^\ell \times 2^\ell$ is defined by induction:

$$H_{2^0} = [1], \quad H_{2^\ell} = \frac{1}{\sqrt{2}} \begin{bmatrix} H_{2^{\ell-1}} & H_{2^{\ell-1}} \\ H_{2^{\ell-1}} & -H_{2^{\ell-1}} \end{bmatrix}$$

- We can assume w.l.o.g. that d is a power of 2 (why?)
- Henceforce we will denote $H := H_d$

- Claims:

- H is an **orthogonal matrix**: $H^T H = H H^T = I$ (proof?)
 - Hence: $\|Hx\|_2 = \|x\|_2$ for all $x \in \mathbb{R}^d$ (proof?)

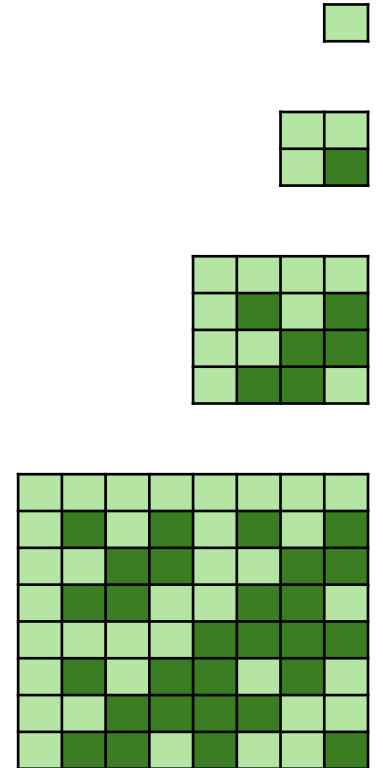


The Hadamard Matrix

- The **Hadamard** matrix of order $2^\ell \times 2^\ell$ is defined by induction:

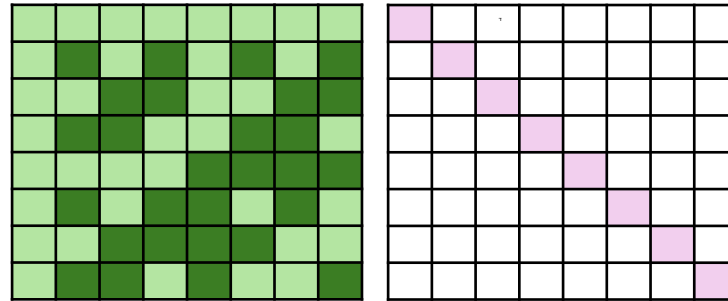
$$H_{2^0} = [1], \quad H_{2^\ell} = \frac{1}{\sqrt{2}} \begin{bmatrix} H_{2^{\ell-1}} & H_{2^{\ell-1}} \\ H_{2^{\ell-1}} & -H_{2^{\ell-1}} \end{bmatrix}$$

- We can assume w.l.o.g. that d is a power of 2 (**why?**)
- Henceforce we will denote $H := H_d$
- Claims:
 - H is an **orthogonal matrix**: $H^T H = H H^T = I$ (**proof?**)
 - Hence: $\|Hx\|_2 = \|x\|_2$ for all $x \in \mathbb{R}^d$ (**proof?**)
 - Given $x \in \mathbb{R}^d$, we can compute Hx in time $O(d \log d)$ (**how?**)



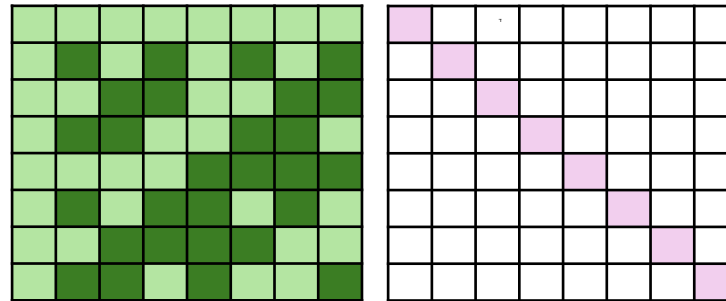
Randomized Hadamard Transform (RHT)

- **RHT:** H times a diagonal matrix D with iid random signs



Randomized Hadamard Transform (RHT)

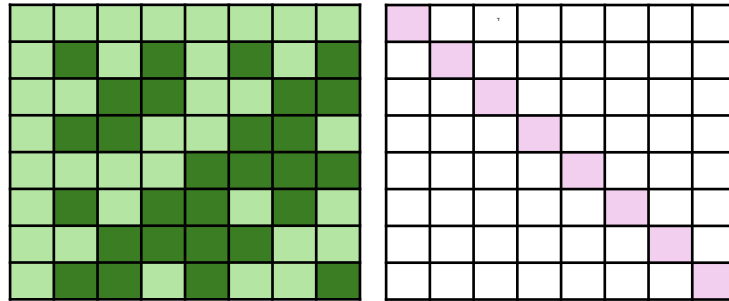
- **RHT:** H times a diagonal matrix D with iid random signs



- **Observe:** Given $x \in \mathbb{R}^d$, computing HDx takes time $O(d \log d)$

Randomized Hadamard Transform (RHT)

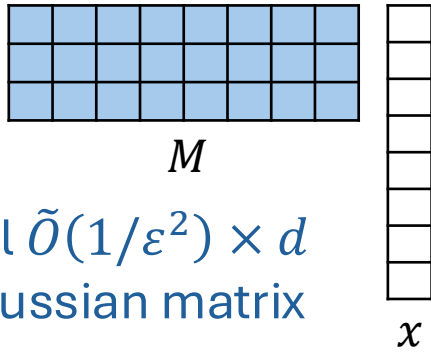
- **RHT:** H times a diagonal matrix D with iid random signs



- **Observe:** Given $x \in \mathbb{R}^d$, computing HDx takes time $O(d \log d)$
- **Lemma (flattening):** HDx has small coordinates
 - We will formalize and prove

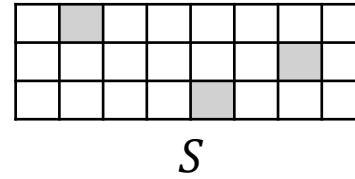
Fast JL via RHT

JL

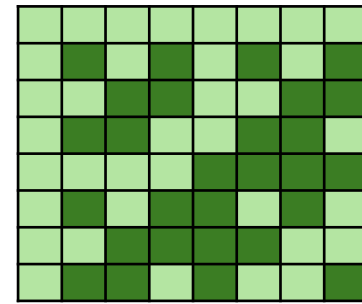


Full $\tilde{O}(1/\varepsilon^2) \times d$
Gaussian matrix

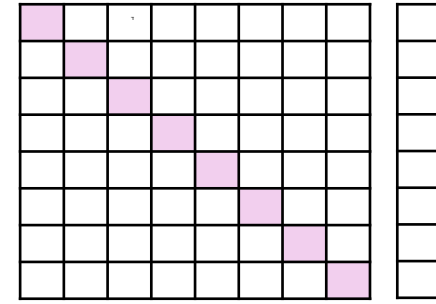
FJLT



$\tilde{O}(1/\varepsilon^2) \times d$
row-sampling
matrix



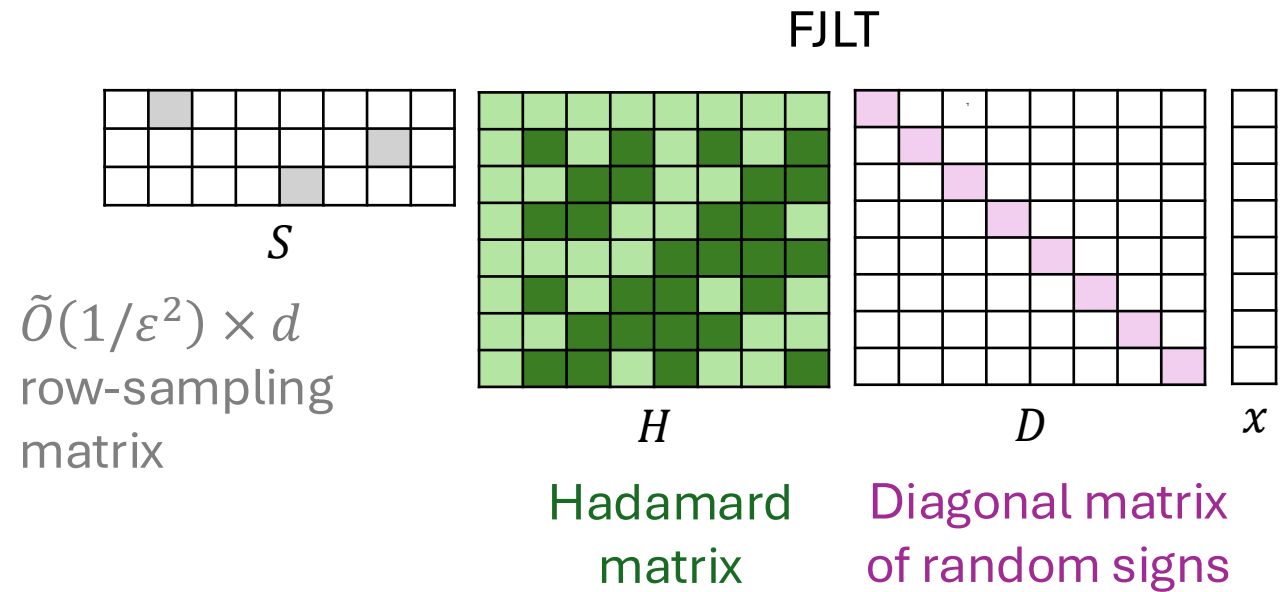
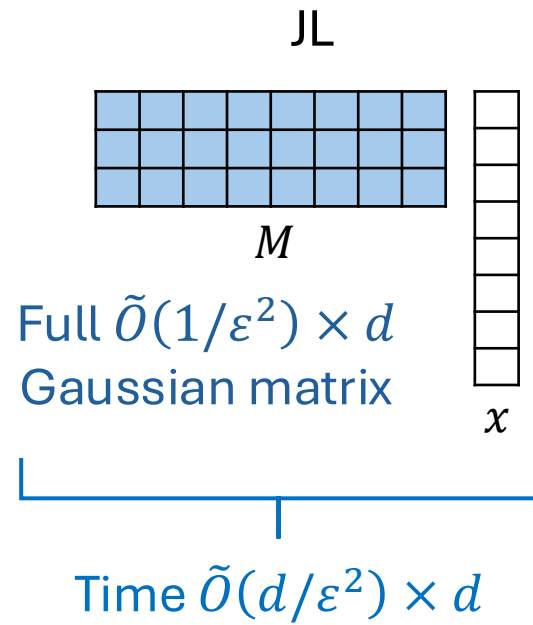
Hadamard
matrix



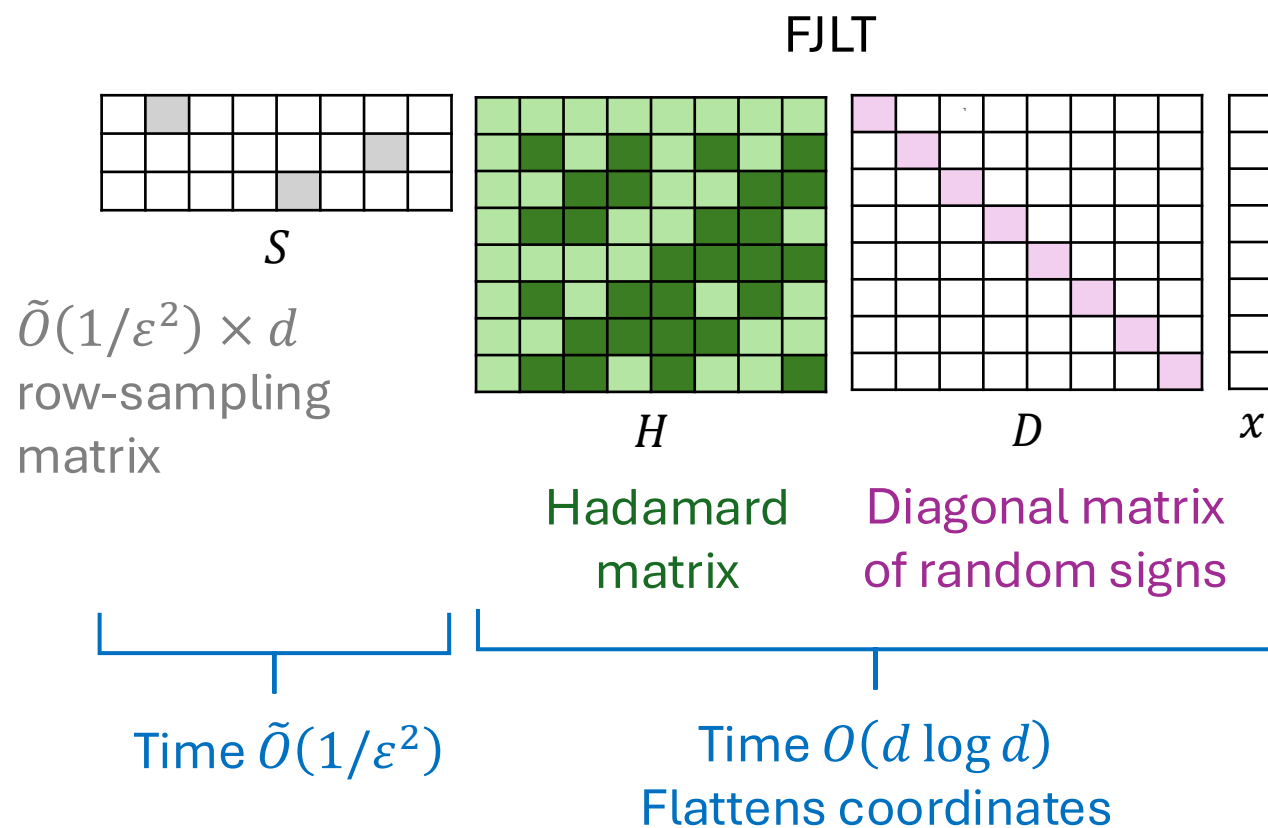
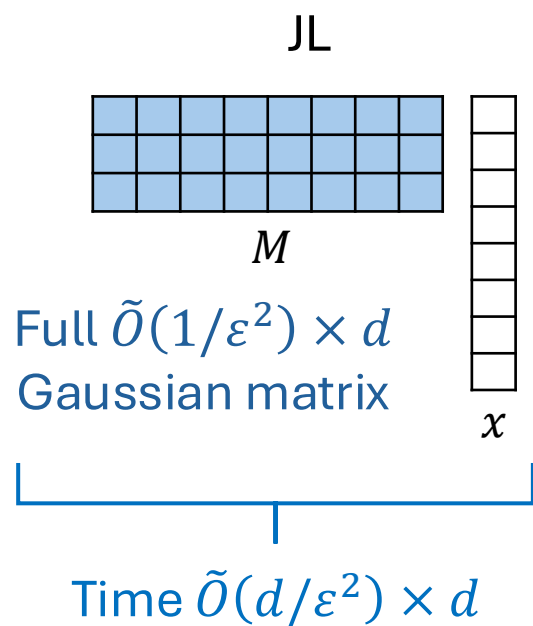
Diagonal matrix
of random signs

x

Fast JL via RHT



Fast JL via RHT



Fast JL via RHT

